



**INSTITUTE OF DATA ENGINEERING, ANALYTICS AND
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CAUSAL CROP YIELD FORECASTING

Using Causal Inference and Machine Learning

Summer Internship Project Report

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Abstract

Crop yield forecasting is an important task for improving agricultural planning, food security, and resource management. Traditional prediction models often rely on statistical correlations, which may overlook the underlying causal relationships between environmental factors and crop yield. This project presents a crop yield forecasting framework that combines causal inference with machine learning and deep learning to improve both prediction accuracy and model interpretability. The study uses the India subset of the CY-Bench (Crop Yield Benchmark) dataset, which contains agro-climatic, soil, vegetation, and environmental variables.

The analysis begins with data preprocessing and exploratory data analysis, followed by the construction of an expert-driven Directed Acyclic Graph (DAG), which is further validated using the NOTEARS causal discovery algorithm. Soil moisture is considered as the treatment variable and crop yield as the outcome variable. The causal effect is estimated using Linear Regression Adjustment, Propensity Score Matching (PSM), Inverse Probability Weighting (IPW), Augmented Inverse Probability Weighting (AIPW), and Double Machine Learning (LinearDML), together with robustness checks using refutation tests.

Based on the causal analysis, nine important features are selected for predictive modelling. Four traditional machine learning models (Linear Regression, Random Forest, XGBoost, and LightGBM) and two deep learning models (Multi-Layer Perceptron and Long Short-Term Memory) are evaluated using both the complete feature set and the selected causal features. Among all the models, XGBoost achieves the best predictive performance with an R^2 score of 0.904 using all features and 0.824 using only causal features, retaining over 91% of its predictive performance. Model predictions are further interpreted using SHAP analysis to identify the most influential factors affecting crop yield. The proposed framework demonstrates that incorporating causal inference into predictive modelling provides accurate, interpretable, and reliable crop yield forecasts that can support data-driven agricultural decision making.

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List of Abbreviations

ATE	Average Treatment Effect
ATT	Average Treatment Effect on Treated
CATE	Conditional Average Treatment Effect
DAG	Directed Acyclic Graph
SCM	Structural Causal Model
PSM	Propensity Score Matching
IPW	Inverse Probability Weighting
AIPW	Augmented Inverse Probability Weighting
RCT	Randomized Controlled Trial
ML	Machine Learning
EDA	Exploratory Data Analysis
NDVI	Normalized Difference Vegetation Index
CCM	Convergent Cross Mapping
LAI	Leaf Area Index
FAO	Food and Agriculture Organization
DoWhy	Causal Inference Library
EconML	Machine Learning for Causal Inference

Chapter 1 Introduction

Agriculture is one of the major sectors of the Indian economy and plays an important role in ensuring food security and supporting the livelihoods of millions of people. However, crop production is influenced by several factors such as weather conditions, soil characteristics, water availability and farming practices. As these factors vary from season to season, estimating crop yield accurately before harvest has become an important problem for farmers, researchers and policymakers.

The availability of large agricultural datasets has enabled the use of data-driven approaches for crop yield prediction. Machine learning models have shown good predictive performance by learning patterns from historical data. However, these models are primarily based on correlations and may assign importance to variables that do not directly influence crop yield. This limits the interpretability of the predictions and makes it difficult to understand the actual factors responsible for changes in yield.

This project combines causal inference and machine learning to improve both prediction accuracy and interpretability of crop yield models. Instead of relying only on statistical relationships, the proposed framework identifies variables that have a direct causal influence on crop yield and uses them for predictive modelling. The study also presents a Streamlit-based decision support application, *Causal Crop Advisor*, which provides yield prediction together with practical recommendations to assist agricultural decision-making.

1.1 Background

Crop yield forecasting has become increasingly important due to its role in agricultural planning and food security. Accurate yield estimates help farmers plan irrigation, fertilizer application and harvesting, while government agencies use such information for procurement, storage and production planning. With changing climatic conditions and increasing uncertainty in agriculture, the demand for reliable forecasting methods has grown

considerably.

Earlier approaches to crop yield estimation mainly relied on field surveys, statistical models and expert knowledge. Although these methods are still useful, they often struggle to capture the complex relationships among weather, soil conditions and crop management practices. The availability of large agricultural datasets, weather records and satellite observations has encouraged the use of data-driven methods for yield prediction.

Machine learning techniques have shown promising results in crop yield forecasting because they can learn complex patterns from historical data. However, most machine learning models are based on correlation rather than cause-and-effect relationships. As a result, they may identify variables that appear important statistically but do not directly influence crop yield. This makes the predictions difficult to interpret and may reduce their reliability under changing environmental conditions.

To overcome this limitation, researchers have started combining causal inference with machine learning. Causal inference focuses on identifying variables that have a direct effect on the outcome while accounting for confounding factors. Integrating causal reasoning with predictive models not only improves interpretability but also provides a better understanding of the factors influencing crop yield.

1.2 Problem Statement

Most existing crop yield forecasting methods rely on statistical correlations to predict agricultural production. Although these approaches often achieve good predictive performance, they provide limited insight into the causal factors influencing crop yield. As a result, the models may become less reliable under changing environmental conditions and offer limited support for practical agricultural decision-making.

This project addresses these limitations by developing a causal crop yield forecasting framework using the CY-Bench dataset. The proposed approach integrates expert-defined causal knowledge with data-driven causal discovery to identify meaningful cause-and-effect relationships. These causal insights are then combined with machine learning models to improve prediction interpretability while supporting an interactive decision support system for agricultural applications.

1.3 Project Relevance

Agriculture is increasingly becoming data-driven, with large amounts of information being generated from weather stations, satellite observations, soil measurements and agricultural surveys. Converting this information into meaningful insights can help improve crop

management and support informed decision-making. Accurate crop yield forecasting is therefore important not only for farmers but also for government agencies involved in production planning, procurement and food security.

Although machine learning has improved the accuracy of crop yield prediction, many existing models behave as black boxes and provide limited explanation for their predictions. In practical agricultural applications, understanding the factors responsible for changes in crop yield is equally important. The integration of causal inference with machine learning addresses this need by identifying variables that have a direct influence on crop production rather than relying only on statistical correlations. The present project contributes towards this objective by combining causal analysis, predictive modelling and an interactive decision support application. The framework provides yield predictions along with information about the factors influencing crop production.

1.4 Technologies Involved

The proposed framework was developed using Python, which served as the primary programming language throughout the project. Data preprocessing, exploratory data analysis and visualization were carried out using libraries such as **Pandas**, **NumPy**, **Matplotlib** and **Seaborn**. These tools were used to clean the dataset, understand variable distributions and prepare the data for further analysis.

The causal analysis was performed using an expert-defined Directed Acyclic Graph (DAG), the NOTEARS causal discovery algorithm, and the **DoWhy** and **EconML** libraries for causal effect estimation. For yield prediction, several machine learning models, including Random Forest, XGBoost, LightGBM, Multi-Layer Perceptron (MLP) and Long Short-Term Memory (LSTM), were implemented and evaluated. Finally, a Streamlit-based application, *Causal Crop Advisor*, was developed to provide crop yield predictions and decision support through an interactive user interface.

1.5 Procedure Followed

The project followed a systematic workflow beginning with data acquisition from the CY-Bench dataset, followed by data cleaning, preprocessing and exploratory data analysis. An expert-defined Directed Acyclic Graph (DAG) was first developed using agricultural domain knowledge and later compared with a data-driven causal graph generated using the NOTEARS algorithm. The identified causal relationships were used for causal effect estimation and feature selection before developing multiple machine learning models for crop yield prediction. Finally, the trained models were integrated into a Streamlit-based decision support application, *Causal Crop Advisor*, to provide crop yield prediction and practical recommendations.

1.6 Training Undertaken During the First Two Weeks

During the initial phase of the internship, a structured training programme was conducted to provide participants with the theoretical foundation and practical skills required for the project. The training covered topics related to statistics, machine learning, deep learning, software development, cloud technologies and modern artificial intelligence. These sessions helped in understanding the concepts that were later applied during the implementation of the project.

Table 1.1: Topics Covered During the First Two Weeks of Internship

S. No.	Training Topic
1	Basic Statistics for Data Science
2	Data Visualization
3	Introduction to GitHub and Cloud Computing
4	Data Science Application Development using Streamlit
5	Machine Learning – I
6	Machine Learning – II
7	LLM Fundamentals
8	Communication Skills
9	Deep Learning – I
10	Deep Learning – II
11	Computer Vision Overview
12	Agentic AI
13	System View of Data Science
14	Reinforcement Learning

1.7 Independent Contribution and Analytical Scope

Along with the primary workflow, a separate analysis pipeline was also explored. So that the pre-processing and causal analysis steps could also be checked independently. This complementary workflow, done in two-stages started from the raw and processed data stage and then moved toward causal exploration using CCM, DAG construction, and DoWhy-based estimation.

Chapter 2 Project Objectives

2.1 Aim of the Project

The primary aim of this project is to develop an interpretable crop yield forecasting framework by integrating causal inference with machine learning techniques. Unlike conventional prediction models that primarily rely on statistical correlations, the proposed framework seeks to identify the causal factors influencing crop yield while maintaining competitive predictive performance. The project also aims to translate the developed models into a practical decision support system through the Streamlit-based application, *Causal Crop Advisor*, enabling users to obtain crop yield predictions together with meaningful agricultural insights.

2.2 Objectives of the Study

The specific objectives of the project are as follows:

1. To develop a causal crop yield forecasting framework using the CY-Bench agricultural dataset.
2. To construct and compare an expert-defined causal graph with a data-driven causal graph generated using the NOTEARS causal discovery algorithm.
3. To estimate the causal effects of important agricultural variables and identify causally significant features using modern causal inference techniques.
4. To evaluate the predictive performance of multiple machine learning models using both the complete feature set and the causally selected feature set.
5. To develop an interactive decision support application (*Causal Crop Advisor*) for crop yield prediction and agricultural decision-making.

2.3 Research Hypothesis

The present study is based on the following research hypotheses:

Null Hypothesis (H_0):

The use of causally selected features does not improve the predictive performance or interpretability of crop yield forecasting models when compared with models trained using the complete feature set.

Alternative Hypothesis (H_1):

The use of causally selected features improves the interpretability of crop yield forecasting models while maintaining comparable or improved predictive performance relative to models trained using the complete feature set.

Chapter 3 Methodology

This chapter presents the methodology adopted to develop the proposed causal crop yield forecasting framework. It describes the complete workflow followed during the project, beginning with data collection and dataset understanding, followed by data preprocessing, exploratory data analysis, causal graph construction, causal effect estimation, predictive modelling and model evaluation. The chapter also discusses the development of the Streamlit-based decision support application, *Causal Crop Advisor*. The implementation of the project was carried out using Python, and the complete source code has been made publicly available through the project’s GitHub repository for reproducibility.

3.1 Research Workflow

The overall methodology adopted in this study is illustrated in Figure 3.1. The project follows a structured pipeline that begins with understanding the CY-Bench dataset and preparing it for analysis. The processed data is then explored through descriptive analysis before constructing an expert-defined causal graph and a data-driven causal graph using the NOTEARS algorithm. The identified causal relationships are subsequently utilized for causal effect estimation and causal feature selection. Finally, the selected features are used for predictive modelling using both traditional machine learning and deep learning approaches, followed by model interpretation and deployment through the *Causal Crop Advisor* decision support application.

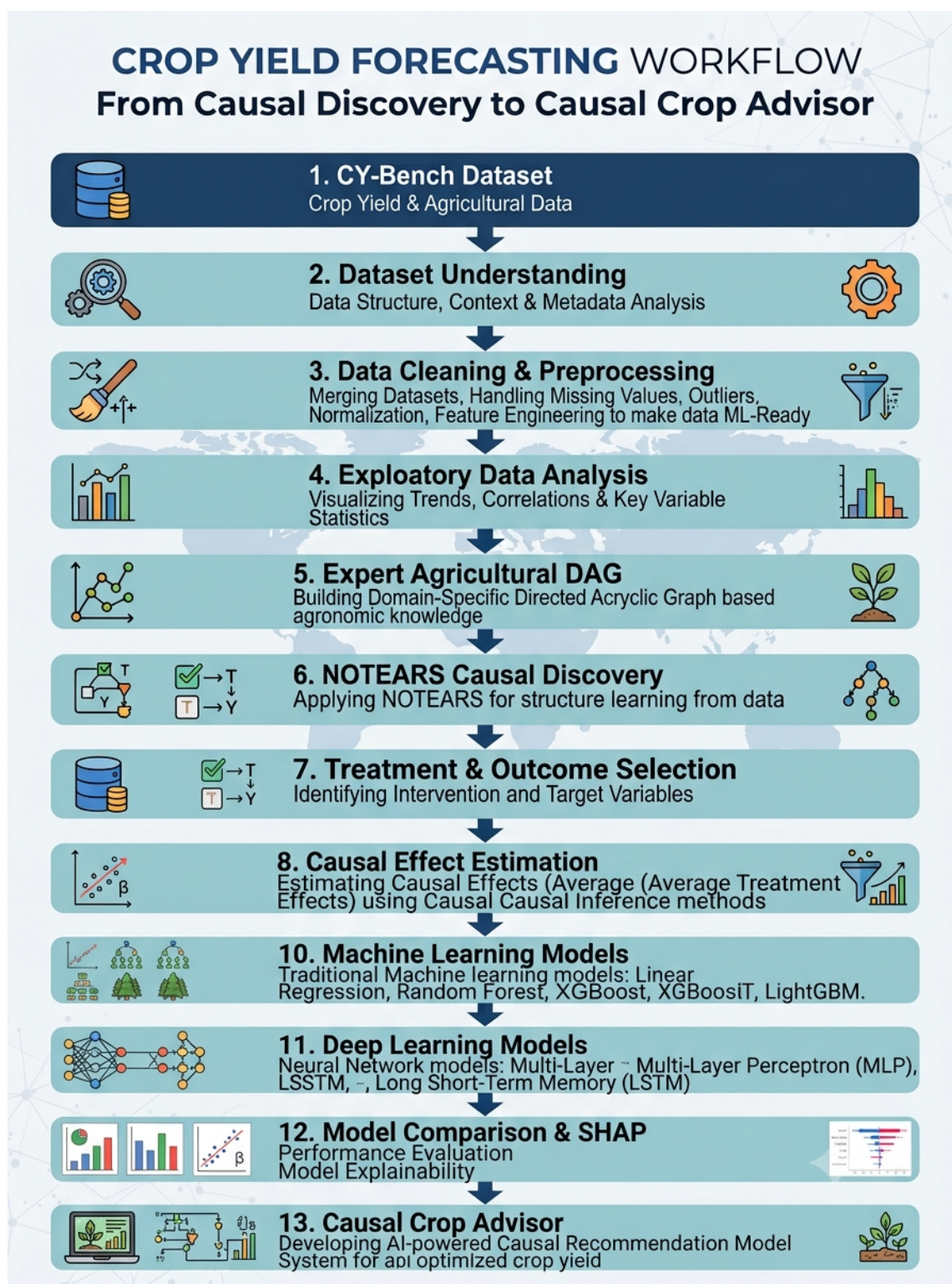


Figure 3.1: Overall workflow adopted for the causal crop yield forecasting framework.

The workflow shown above summarizes the sequence of activities performed throughout the project. Each stage is discussed in detail in the subsequent sections of this chapter, covering data preparation, causal modelling, predictive modelling, model evaluation and application development.

3.2 Dataset Description and Data Collection

The study was carried out using the **CY-Bench** benchmark dataset, a publicly available dataset developed for crop yield forecasting research. For this project, only the **Indian wheat subset** of the dataset was considered. CY-Bench provides multiple data sources describing agricultural production, weather conditions, soil characteristics, vegetation indices and geographical information. Combining these heterogeneous datasets makes it possible to analyse both environmental and management factors influencing crop yield.

The dataset consists of historical observations collected at the administrative-region level across different years. Yield records are linked with weather variables, soil information, crop calendar data, remotely sensed vegetation indices and soil moisture observations through a common administrative identifier (`adm_id`). This integrated structure makes the dataset suitable for both causal inference and machine learning applications.

The datasets used in this project are summarised in Table 3.1. The yield dataset served as the primary dataset, while the remaining datasets were used to provide explanatory variables describing climatic, geographical and environmental conditions.

Table 3.1: CY-Bench Dataset Components Used in the Study

Dataset	Description	Records
Yield	Historical wheat yield observations	11,706
Weather	Daily meteorological variables including temperature, precipitation and radiation	4,821,600
Soil	Static soil characteristics	574
Soil Moisture	Surface and root-zone soil moisture measurements	4,384,212
NDVI	Normalized Difference Vegetation Index	530,395
FPAR	Fraction of Photosynthetically Active Radiation	475,272
Location	Latitude, longitude and regional area information	574
Crop Calendar	Crop growing season information	574
Crop Mask	Crop cultivation area and coverage	585

The primary yield dataset contains historical wheat production information for different administrative regions in India. It consists of 11,706 observations and 12 variables covering harvest year, production, harvested area and crop yield. During the initial assessment of the dataset, it was observed that a few variables related to planting information were completely unavailable and therefore were excluded during preprocessing.

A summary of the primary yield dataset is presented in Table 3.2.

Table 3.2: Summary of the Primary Yield Dataset

Attribute	Value
Crop	Wheat
Country	India
Number of Records	11,706
Number of Variables	12
Target Variable	Yield
Spatial Unit	Administrative Region (<code>adm_id</code>)
Study Period	1990–2017

The variables considered during modelling are listed in Table 3.3. These variables collectively represent crop production, weather conditions, soil properties and remotely sensed indicators, providing a comprehensive description of the agricultural environment.

Table 3.3: Major Variables Used for Crop Yield Forecasting

Variable	Description
Yield	Crop yield (target variable)
Production	Total wheat production
Harvest Area	Harvested agricultural area
Temperature	Minimum, maximum and average temperature
Precipitation	Daily rainfall observations
Solar Radiation	Incoming solar radiation
Soil Moisture	Surface and root-zone moisture
NDVI	Vegetation health indicator
FPAR	Photosynthetic activity indicator
Soil Characteristics	Available water capacity, bulk density and drainage class

3.3 Data Preprocessing and Feature Engineering

Raw datasets obtained from the CY-Bench repository required extensive preprocessing before they could be used for causal analysis. Since the datasets originated from different sources and possessed varying temporal resolutions, a systematic preprocessing pipeline was developed to ensure consistency across all data sources.

Loading and Initial Inspection

All datasets corresponding to Indian wheat production were imported into Python using the Pandas library. The following datasets were loaded:

- Yield records
- Daily weather observations
- Soil properties
- Soil moisture measurements
- NDVI vegetation index
- FPAR vegetation index
- Geographic location information
- Crop calendar
- Crop mask information

Each dataset was inspected to verify its dimensions, data types and overall structure before further processing.

Date Conversion and Time Feature Extraction

The weather, soil moisture, NDVI and FPAR datasets contained date values in integer format (YYYYMMDD). These values were converted into Python datetime objects to facilitate temporal analysis.

From each date, the following temporal features were extracted:

- Year
- Month
- Day of Year (DOY)

These features enabled efficient filtering of observations corresponding to the crop growing season.

Temporal Alignment

Since the datasets covered different time periods, only observations overlapping with the available yield records were retained.

The yield dataset was filtered to include harvest years from 2003 to 2017, ensuring consistency with the availability of satellite and weather observations.

Integration of Static Agricultural Information

The yield dataset was merged with multiple static datasets using the district identifier (`adm_id`) as the common key.

The merged information included:

- Crop calendar (Start of Season and End of Season)
- Soil characteristics
- Geographic coordinates
- Regional area
- Crop area statistics

These variables provide location-specific agricultural characteristics that remain relatively constant over time.

Seasonal Feature Engineering

The CY-Bench datasets contain daily observations, whereas crop yield is reported annually. Therefore, daily observations were aggregated into seasonal summaries based on the crop growing period.

Using the crop calendar, the complete wheat season was identified for each district by combining observations from:

- the previous year's Start of Season (SOS), and
- the current year's End of Season (EOS).

A generic aggregation function was implemented to compute seasonal statistics for every yield observation.

The following seasonal features were generated:

- Average minimum temperature
- Average maximum temperature

- Average mean temperature
- Average solar radiation
- Average evapotranspiration
- Average vapor pressure deficit
- Average climatic water balance
- Average surface soil moisture
- Average root zone soil moisture
- Average NDVI
- Average FPAR

These aggregated variables summarize environmental conditions experienced throughout the crop growth period.

Handling Missing Values

Missing values primarily originated from districts for which crop calendar or static geographical information was unavailable.

Columns containing no useful information (such as planting dates and harvest dates) were removed from the dataset. Records with incomplete seasonal information were subsequently discarded to ensure that all observations contained complete feature vectors.

Duplicate observations were also checked and removed where necessary.

Final Processed Dataset

The fully processed dataset was exported as a CSV file for subsequent exploratory data analysis, causal graph construction and causal effect estimation.

The final dataset consisted of 6,715 complete district-year observations and 28 variables representing crop yield, static agricultural information and engineered seasonal environmental features.

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was carried out to understand the characteristics of the processed crop yield dataset before causal modelling. The analysis included examination of feature distributions, summary statistics, correlation analysis, temporal trends, spatial variation, and relationships between crop yield and important explanatory variables.

The objective of EDA was to identify potential data quality issues, understand variable behaviour, and provide an initial understanding of the dataset before constructing the causal framework.

The analyses performed included:

- Distribution analysis of crop yield
- Summary statistics of numerical variables
- Pearson correlation analysis
- Temporal trend analysis
- Spatial distribution analysis
- Scatter plot analysis between crop yield and important predictor variables

The findings obtained from these analyses are presented and discussed in Chapter 5.

3.5 Causal Inference Framework

Traditional machine learning models primarily capture statistical associations and often fail to distinguish causal relationships from simple correlations. In agricultural systems, crop yield is influenced by multiple interconnected factors such as weather conditions, soil properties, vegetation indices, and management practices, making causal analysis essential for identifying the true drivers of crop productivity.

To address this challenge, a causal inference framework was developed that combines expert agricultural knowledge with data-driven causal discovery techniques. The framework aims to estimate the direct causal effects of selected treatment variables on crop yield while accounting for potential confounding factors. The resulting causal information is further utilized for feature selection and predictive modelling, improving both model interpretability and prediction performance.

The overall causal inference pipeline consists of the following stages:

- Construction of an expert agricultural Directed Acyclic Graph (DAG)
- Data-driven causal discovery using the NOTEARS algorithm
- Selection of treatment, outcome, and confounding variables
- Estimation of causal effects using multiple causal inference methods
- Selection of causally relevant features for predictive modelling

3.5.1 Expert Agricultural Directed Acyclic Graph (DAG)

Before applying data-driven causal discovery techniques, an expert-defined Directed Acyclic Graph (DAG) was constructed to represent the assumed causal relationships among the variables influencing crop yield. The DAG was designed using established agricultural knowledge and findings from previous research on crop production, climate interactions, soil characteristics, and vegetation dynamics.

A Directed Acyclic Graph is a graphical representation in which each node corresponds to a variable and each directed edge represents a hypothesized causal relationship. Since the graph contains no directed cycles, it provides a valid causal structure that can be used for identifying confounding variables, treatment variables, mediators, and outcome variables required for causal effect estimation.

In this study, the expert DAG was developed prior to performing data-driven causal discovery. This ensures that domain expertise is incorporated into the causal modelling process and serves as a baseline for comparing the causal relationships learned automatically by the NOTEARS algorithm in the subsequent section.

The variables included in the study were grouped into six major categories according to their roles in the agricultural system:

- **Geographical Variable:** Latitude
- **Temporal Variable:** Harvest Year
- **Climate Variables:** Average Temperature, Solar Radiation and Climate Water Balance
- **Soil Variables:** Available Water Capacity, Bulk Density and Drainage Class
- **Treatment Variable:** Average Surface Soil Moisture (SSM)
- **Mediator Variables:** NDVI and FPAR
- **Outcome Variable:** Crop Yield

The selection of these variables was based on agronomic principles and their known influence on crop growth and productivity.

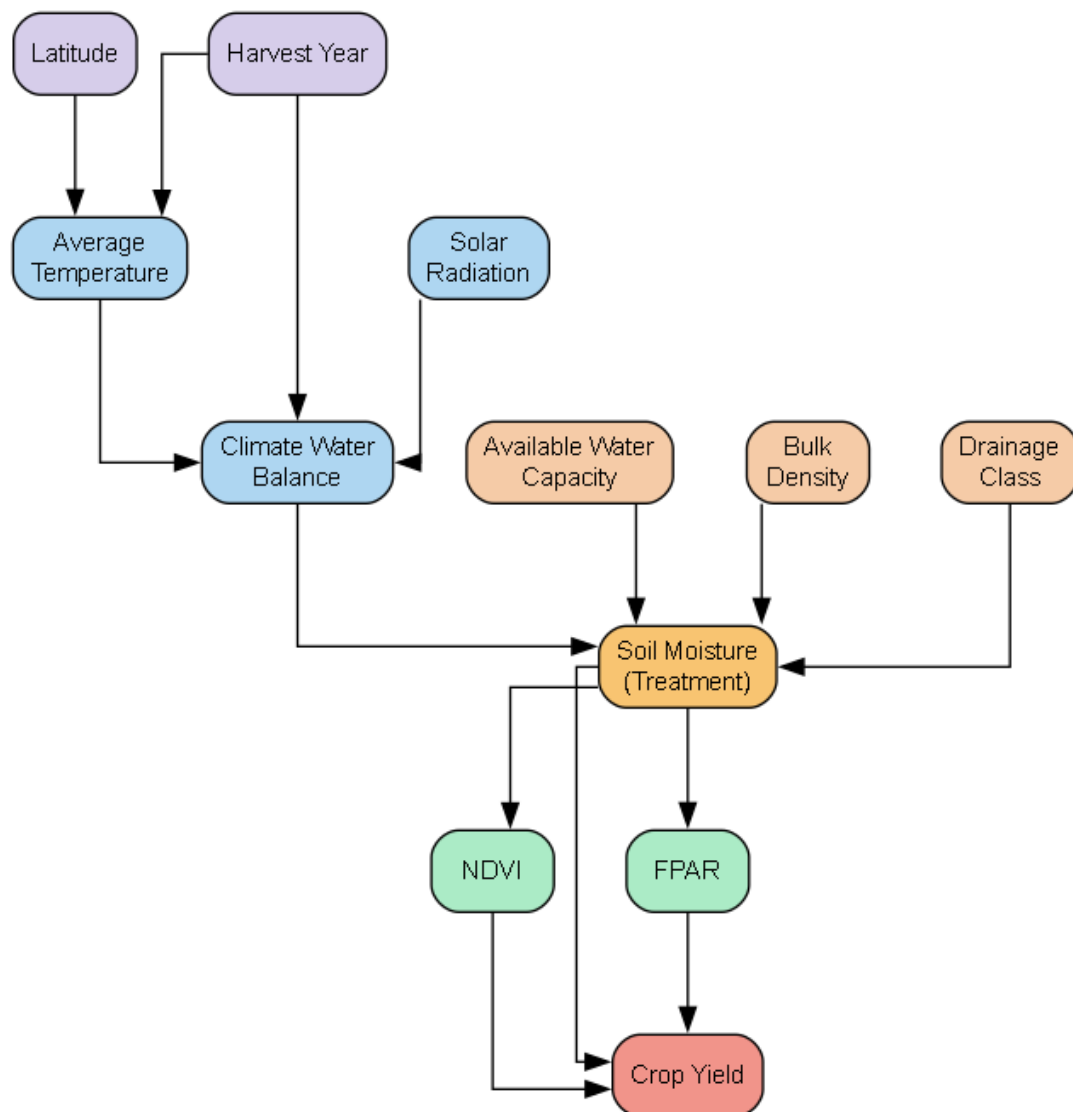


Figure 3.2: Agriculture domain Knowledge-Based Agricultural Directed Acyclic Graph showing the assumed causal relationships among geographical, climatic, soil, vegetation and crop yield variables.

Data-driven Causal Discovery

Although the expert agricultural DAG incorporates domain knowledge, it may not capture all statistical relationships present in the observed data. Therefore, data-driven causal discovery techniques were applied to automatically infer the dependency structure among the selected variables. These algorithms estimate causal relationships directly from the dataset and provide an empirical graph that can be compared with the expert-defined causal framework.

Two complementary causal discovery approaches were employed in this study. The first is the PC (Peter-Clark) algorithm, a constraint-based method that identifies causal structures through conditional independence tests. The second is the NOTEARS algorithm, an optimization-based approach that formulates causal structure learning as a

continuous optimization problem while enforcing the acyclic property of directed graphs.

The learned causal graphs were not used directly for causal effect estimation. Instead, they served as validation tools for assessing the consistency of the expert agricultural DAG and identifying statistically supported relationships within the dataset.

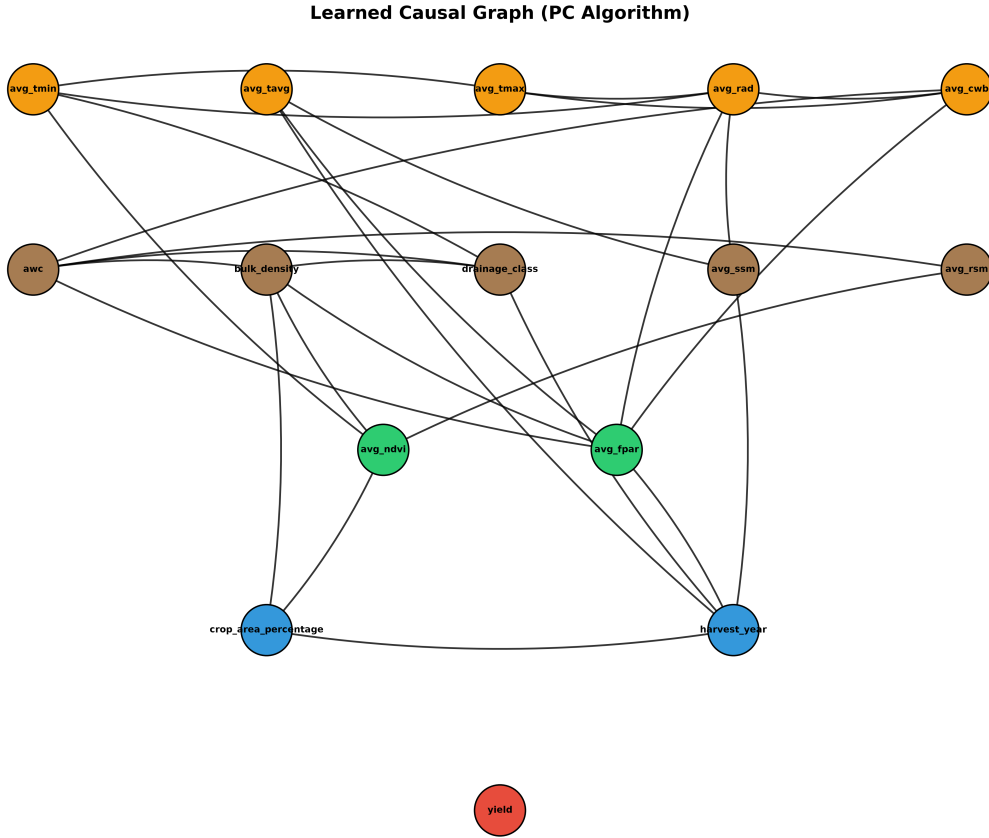


Figure 3.3: Causal graph learned using the PC algorithm.

The PC algorithm identified a directed acyclic graph consisting of 15 variables connected through 29 statistically inferred causal relationships. The resulting graph satisfied the acyclic property, confirming its suitability as a causal graph.

Several relationships discovered by the PC algorithm were consistent with agricultural reasoning, including the influence of climatic variables on soil moisture and vegetation indices. Variables such as *avg_fpar*, *avg_rad*, and *awc* exhibited relatively high connectivity, indicating their importance within the learned network.

However, some expected agronomic relationships were not recovered. In particular, no direct parent of crop yield was identified, despite soil moisture and vegetation indices being known determinants of agricultural productivity. This limitation highlights that purely data-driven causal discovery may overlook meaningful causal pathways because of finite sample effects, statistical assumptions, or hidden confounding variables.

NOTEARS Algorithm To complement the constraint-based PC algorithm, the NOTEARS (Non-combinatorial Optimization via Trace Exponential and Augmented lagRangian for Structure learning) algorithm was employed. Unlike traditional search-based methods, NOTEARS formulates causal structure learning as a continuous optimization problem while explicitly enforcing the directed acyclic graph (DAG) constraint. This approach enables efficient estimation of causal relationships by optimizing an adjacency matrix rather than searching through all possible graph structures.

The standardized dataset containing the selected agricultural, climatic, soil, vegetation, and temporal variables was used as the input to the NOTEARS algorithm. The resulting adjacency matrix was converted into a directed graph, from which the learned causal relationships were extracted and visualized for comparison with the expert-defined DAG.

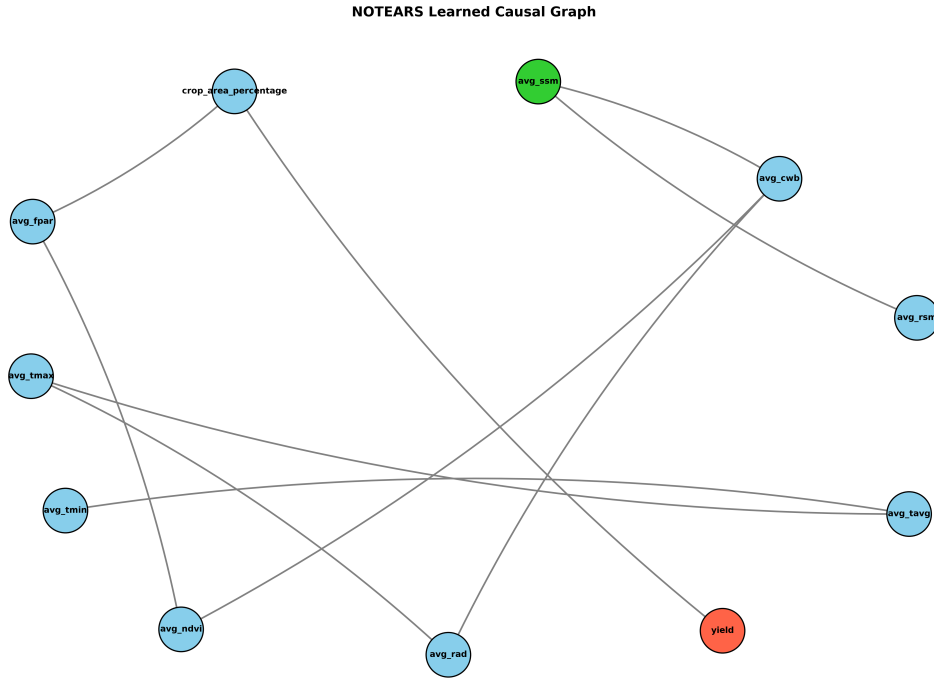


Figure 3.4: Directed Acyclic Graph learned using the NOTEARS algorithm.

The NOTEARS algorithm produced a comparatively sparse causal graph consisting of 11 variables connected through 10 directed relationships. The learned graph satisfied the acyclic property and represented the most statistically significant dependencies identified from the data.

Several discovered relationships were consistent with agricultural understanding. For example, solar radiation influenced maximum temperature, which subsequently affected average temperature. Climate water balance was identified as an important determinant of surface soil moisture, while surface soil moisture influenced root-zone soil moisture. In addition, NDVI was found to affect both FPAR and climate water balance, indicating

strong interactions among vegetation-related variables.

Despite recovering several meaningful environmental relationships, the learned graph did not identify a direct causal effect of surface soil moisture on crop yield. Instead, crop area percentage emerged as the only direct parent of yield. Since this relationship is not fully consistent with established agronomic knowledge, the NOTEARS graph was considered primarily as a data-driven validation tool rather than the causal graph used for downstream effect estimation.

Comparison of Expert and Learned Causal Graphs The causal structures obtained from the Expert DAG, the PC algorithm, and the NOTEARS algorithm were compared to evaluate the consistency between domain knowledge and statistically learned relationships. While the expert DAG was constructed using established agronomic principles and findings from previous literature, the PC algorithm and NOTEARS inferred causal relationships directly from the observed data.

Table 3.4 summarizes the major differences among the three approaches.

Table 3.4: Comparison of the Expert DAG, PC Algorithm and NOTEARS

Criterion	Expert DAG	PC Algorithm	NOTEARS
Construction Method	Domain Knowledge	Constraint-based Learning	Optimization-based Learning
Number of Variables	12	15	11
Number of Relationships	14	29	10
Basis of Learning	Agricultural Expertise	Conditional Independence Tests	Continuous Optimization
Suitable for Effect Estimation	Yes	No	No
Purpose in this Study	Final Causal Model	Validation	Validation

The PC algorithm identified a relatively dense causal network containing 29 relationships. Although several statistically significant dependencies were discovered, some important agronomic relationships, including the direct influence of soil moisture on crop yield, were not recovered. Similarly, the NOTEARS algorithm produced a sparse causal graph that successfully identified meaningful environmental relationships, such as the influence of climate water balance on soil moisture, but also failed to recover several expected causal pathways supported by agricultural knowledge.

These differences arise because data-driven causal discovery algorithms rely solely on statistical dependencies present in the observed data. Consequently, they may overlook genuine causal mechanisms due to limited sample size, measurement noise, hidden confounders, or violations of their underlying assumptions.

Therefore, the expert agricultural DAG was selected as the final causal framework for subsequent causal effect estimation using DoWhy and EconML. The learned causal graphs obtained from the PC algorithm and NOTEARS were retained as validation tools to assess the consistency of the assumed causal structure rather than as the primary causal model.

3.5.2 Treatment and Outcome Selection

Following the construction of the expert causal framework and the comparison with the learned causal graphs, the treatment, outcome, confounders, and mediators were identified based on both domain knowledge and the assumed causal relationships represented in the expert DAG. Selecting these variables appropriately is essential for obtaining an unbiased estimate of the treatment effect during causal inference.

The treatment variable selected in this study is the **Average Surface Soil Moisture (avg_ssm)**, while the response variable is **Crop Yield (yield)**. Surface soil moisture was chosen as the treatment because it directly influences crop growth and can be affected by several climatic and soil-related factors, making it an appropriate intervention variable for estimating causal effects on agricultural productivity.

The outcome variable is crop yield, which represents the final productivity of a crop and serves as the primary variable of interest in this study.

Based on the expert agricultural DAG, several variables were identified as potential confounders since they influence both soil moisture and crop yield. These include average temperature, solar radiation, climate water balance, available water capacity, bulk density, drainage class, and harvest year. Adjusting for these variables helps satisfy the backdoor criterion and reduces confounding bias during causal effect estimation.

Vegetation indices, namely **NDVI** and **FPAR**, were treated as mediator variables because they represent intermediate biological processes through which soil moisture influences crop growth before ultimately affecting crop yield.

The selected variables used throughout the causal inference analysis are summarized in Table 3.5.

Table 3.5: Treatment, outcome, confounders and mediator variables used for causal inference.

Category	Variable(s)
Treatment	Average Surface Soil Moisture (avg_ssm)
Outcome	Crop Yield (yield)
Confounders	Average Temperature (avg_tavg), Solar Radiation (avg_rad), Climate Water Balance (avg_cwb), Available Water Capacity (awc), Bulk Density (bulk_density), Drainage Class (drainage_class), Harvest Year (harvest_year)
Mediators	Normalized Difference Vegetation Index (avg_ndvi), Fraction of Photosynthetically Active Radiation (avg_fpar)

3.5.3 Causal Effect Estimation

Objective

The objective of this stage was to estimate the causal effect of average surface soil moisture (avg_ssm) on crop yield while controlling for potential confounding variables. Unlike traditional predictive models, causal effect estimation aims to quantify the direct impact of soil moisture on crop yield after adjusting for environmental and soil-related factors.

Treatment, Outcome and Confounders

The causal inference framework was defined using the variables shown in Table 3.6.

Table 3.6: Treatment, Outcome and Covariates

Component	Variable(s)
Treatment	avg_ssm
Outcome	yield
Confounders	avg_tavg, avg_rad, avg_cwb, awc, bulk_density, drainage_class, harvest_year
Mediators	avg_ndvi, avg_fpar

The confounding variables were selected using the Expert Directed Acyclic Graph (DAG) developed during the causal discovery stage.

Identification of the Causal Effect

The causal model was constructed using the DoWhy framework. The backdoor criterion was satisfied using the selected confounders, allowing identification of the Average Treatment Effect (ATE). Under the assumption of no unobserved confounding, the identified

estimand provides a valid estimate of the causal relationship between soil moisture and crop yield.

Linear Regression Adjustment

The first causal estimate was obtained using backdoor linear regression adjustment.

The estimated Average Treatment Effect (ATE) was

$$ATE = -0.2414$$

indicating a negative causal relationship between surface soil moisture and crop yield after adjusting for confounding variables.

Propensity Score Matching

A binary treatment variable was created by splitting observations based on the median value of average surface soil moisture.

Propensity scores were estimated using Logistic Regression, followed by one-to-one nearest neighbour matching to construct balanced treated and control groups.

The estimated treatment effect was

$$ATE = -0.7072$$

Inverse Probability Weighting

Stabilized inverse probability weights were computed using the estimated propensity scores.

The weighted average treatment effect was estimated as

$$ATE = -0.4364$$

This method assigns larger weights to underrepresented observations in order to approximate a randomized experiment.

Double Machine Learning (LinearDML)

To obtain a more robust estimate, Linear Double Machine Learning (LinearDML) from the EconML framework was employed.

Random Forest models were used for both the treatment model and the outcome model.

The estimated Average Treatment Effect was

$$ATE = -0.1192$$

This approach reduces estimation bias while allowing flexible machine learning models to capture nonlinear relationships.

Comparison of Causal Estimates

Table 3.7 compares the estimated treatment effects obtained using different causal inference techniques.

Table 3.7: Comparison of Average Treatment Effects

Method	ATE
Linear Regression	-0.2414
Propensity Score Matching	-0.7072
Inverse Probability Weighting	-0.4364
LinearDML	-0.1192

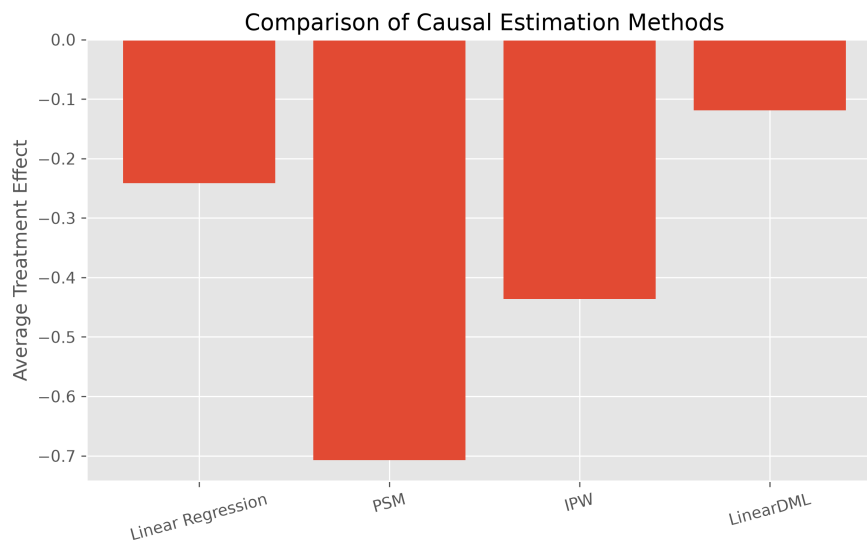


Figure 3.5: Comparison of Average Treatment Effects Estimated Using Different Methods

All estimation methods consistently produced a negative Average Treatment Effect, suggesting that higher surface soil moisture was associated with lower crop yield under the assumptions of the specified causal model.

Model Validation Using Refutation Tests

Three refutation tests were conducted using the DoWhy framework to evaluate the robustness of the estimated causal effect.

Table 3.8: Refutation Test Results

Refutation Test	New Effect	p-value
Random Common Cause	-0.2414	0.90
Placebo Treatment	≈ 0	0.00
Data Subset Refuter	-0.2416	0.98

The random common cause and data subset refutation tests produced estimates very similar to the original causal estimate, indicating strong robustness. The placebo treatment test produced an effect close to zero, confirming that the original estimate was not due to random chance.

Individual Treatment Effects

LinearDML was also used to estimate Individual Treatment Effects (ITE), capturing heterogeneity in treatment effects across observations.

The summary statistics are presented in Table 3.9.

Table 3.9: Summary of Individual Treatment Effects

Statistic	Value
Mean	-0.1192
Standard Deviation	0.1222
Minimum	-0.7494
Maximum	0.3067

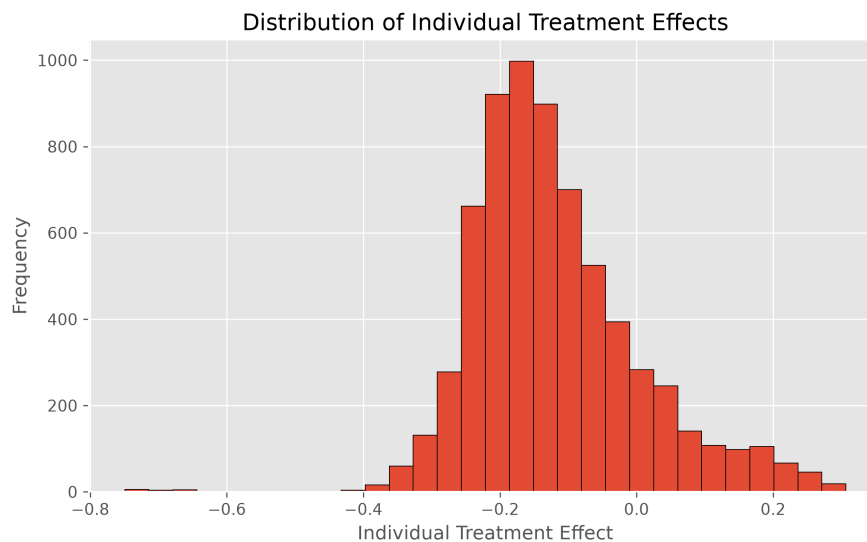


Figure 3.6: Distribution of Individual Treatment Effects

The distribution indicates that the treatment effect varies across observations, suggesting heterogeneous responses of crop yield to soil moisture under different environmental and soil conditions.

Summary

Four causal inference techniques were implemented to estimate the causal effect of average surface soil moisture on crop yield. Although the magnitude of the estimated effects varied across methods, all approaches consistently indicated a negative causal relationship. Robustness was further verified through multiple refutation tests, providing confidence in the validity of the estimated causal effects. These causal estimates form the basis for the subsequent feature selection and predictive modelling stage.

3.5.4 Causal Feature Selection

After estimating the causal effects, the next step was to identify the variables that have the greatest causal relevance for crop yield prediction. Instead of selecting variables solely on the basis of statistical correlation, feature selection was performed by combining information from the expert-defined causal graph, the data-driven NOTEARS graph, and the estimated causal effects obtained in the previous stage. This approach ensures that only variables with meaningful causal relationships are retained for predictive modeling.

Initially, the processed dataset contained 28 variables. The crop yield was selected as the target variable, while all remaining variables were evaluated for their causal importance. Seven variables were identified as causal predictors based on domain knowledge and causal discovery.

Table 3.10: Selected Causal Features

Feature	Description
avg_ssm	Average surface soil moisture
avg_tavg	Average temperature
avg_rad	Average solar radiation
avg_cwb	Average climatic water balance
awc	Available water capacity
bulk_density	Soil bulk density
drainage_class	Soil drainage class

To prioritize the selected variables, a causal importance score was computed for each feature. The score combined three sources of information:

- Connectivity in the expert-defined causal DAG.
- Connectivity in the NOTEARS causal graph.
- Direct causal contribution observed through the LinearDML treatment effect estimation.

Table 3.11 presents the resulting causal feature ranking.

Table 3.11: Causal Feature Ranking

Feature	Expert DAG	NOTEARS	Causal Effect	Total Score
avg_ssm	6	3	3	12
avg_tavg	4	4	0	8
avg_cwb	4	4	0	8
avg_rad	3	2	0	5
awc	1	0	0	1
bulk_density	1	0	0	1
drainage_class	1	0	0	1

Figure 3.7 illustrates the overall causal importance scores. Surface soil moisture received the highest score, indicating that it is the most influential causal factor affecting crop yield. Temperature and climatic water balance were ranked next, highlighting the dominant influence of environmental conditions on agricultural productivity.

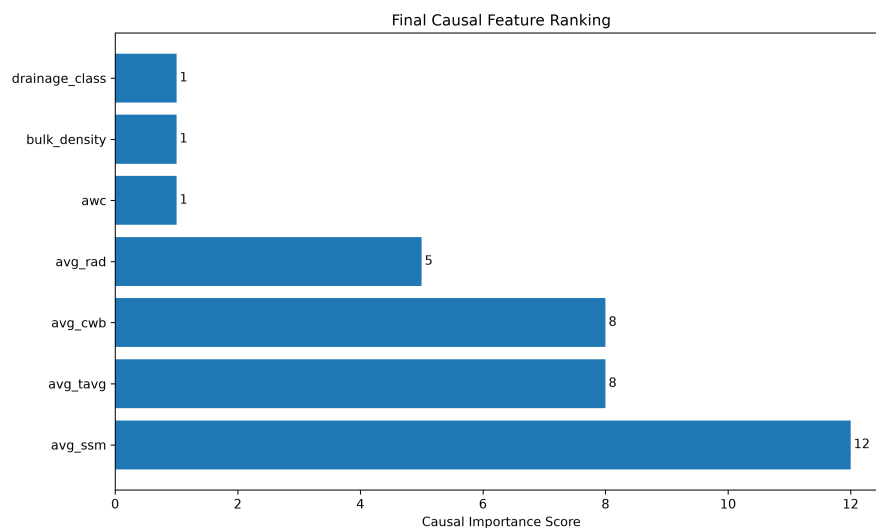


Figure 3.7: Causal Feature Ranking Based on Combined Importance Score

To understand the statistical relationship between the selected variables and crop yield, Pearson correlation coefficients were computed. Figure 3.8 presents the correlations between the selected causal features and the target variable.

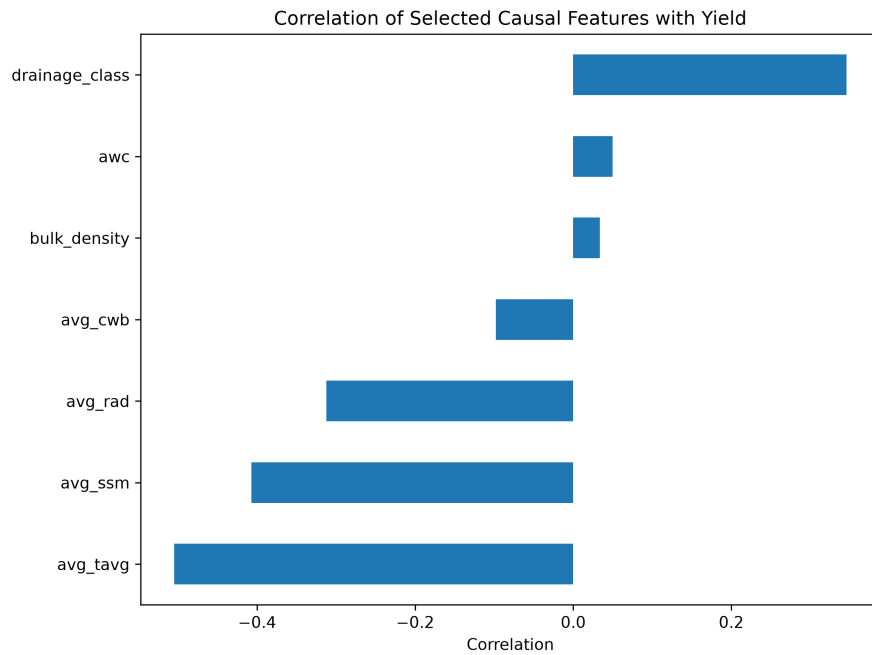


Figure 3.8: Correlation Between Selected Causal Features and Crop Yield

The relationships among the selected variables were further examined using a correlation heatmap shown in Figure 3.9. The heatmap confirms that although several variables exhibit moderate correlations, each contributes distinct information relevant to crop yield prediction.

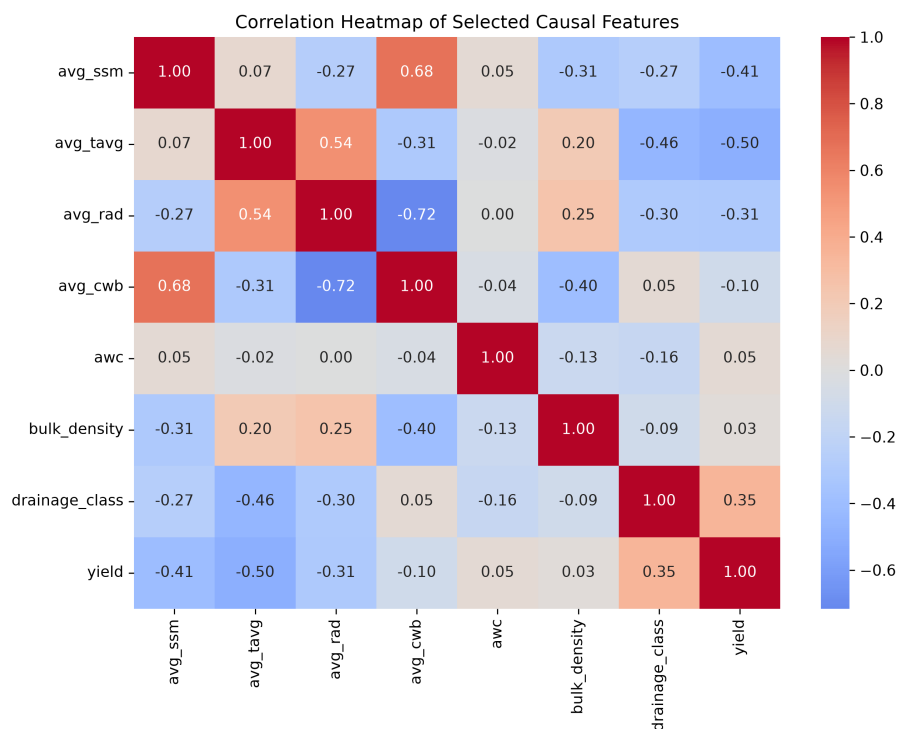


Figure 3.9: Correlation Heatmap of Selected Causal Features

Finally, the complete feature selection workflow is illustrated in Figure 3.10. Starting from the original dataset of 28 variables, causal knowledge from the expert DAG, NOTEARS, and causal effect estimation was integrated to identify seven causally relevant features for the subsequent machine learning models.

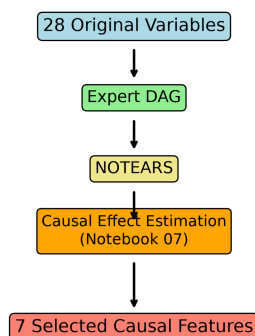


Figure 3.10: Causal Feature Selection Pipeline

The resulting feature set balances domain expertise and data-driven evidence, ensuring that the predictive models rely on variables with genuine causal relevance rather than spurious statistical associations. This improves both model interpretability and robustness while reducing unnecessary predictor variables.

3.5.5 Complementary Analytical Approach

First Run: CCM on Prepared Data

For the first analysis run, work was conducted on the panel dataset used in the primary workflow. The data set contained 6,715 observations and 28 columns, including yield, production, harvest area, soil properties, geographic variables, and aggregated climate and vegetation indicators.

The first run focused on:

- Data checking.
- Panel cleaning.
- Preparation of pairwise CCM inputs.
- Execution of CCM on selected variable pairs.

The CCM preparation stage:

- standardized columns.
- sorted the panel by administrative unit and harvest year.
- applied minimum temporal coverage rules.
- created pair tables for variables such as soil moisture with NDVI, FPAR, and yield.
- prepared climate-water-balance relationships.

The purpose of this stage was to work with the previously prepared data and expert-informed DAG in order to carry out the first CCM exploration and understand how the proposed relationships behaved under nonlinear time-series analysis.

Independent Pre-processing and second CCM run

For the pre-processing, raw CY-Bench wheat dataset for India was assembled from multiple source tables. Namely:

- Crop Calendar
- Crop Mask
- FPAR
- Location
- Meteorology
- NDVI
- Soil Moisture
- Soil
- Yield

First table dimensions, schema, data types, and missingness were inspected. Then key identifiers such as `crop_name`, `adm_id`, `country_code` and converted date fields were standardised into a consistent format before integration.

Some irrelevant or fully empty yield-side columns were dropped and static and dynamic sources were combined into a panel-style dataset.

For the second analysis run, pre-processing was performed independently using the raw CY-Bench wheat India files. These source tables included crop calendar, crop mask,

FPAR, location, meteorology, NDVI, soil moisture, soil, and yield data, with substantial variation in size across tables.

The panel data was then sorted using `adm_id` and harvest year. Then split into dynamic variables and static covariates. 30 relevant columns were retained. And variables such as yield, production, soil moisture *ssm*, NDVI, FPAR, temperature, precipitation, VPD, ET0, and climatic water balance as dynamic signals for time-series investigation.

Then selected variable pairs were:

- converted into CCM-ready tables,
- standardised using z-scored signal columns,
- filtered to keep units with sufficient observations,
- and assigned a local time index within each unit.

Then CCM was run using `pyEDM` to explore directional dependence among selected variable pairs.

DAG construction and the DoWhy

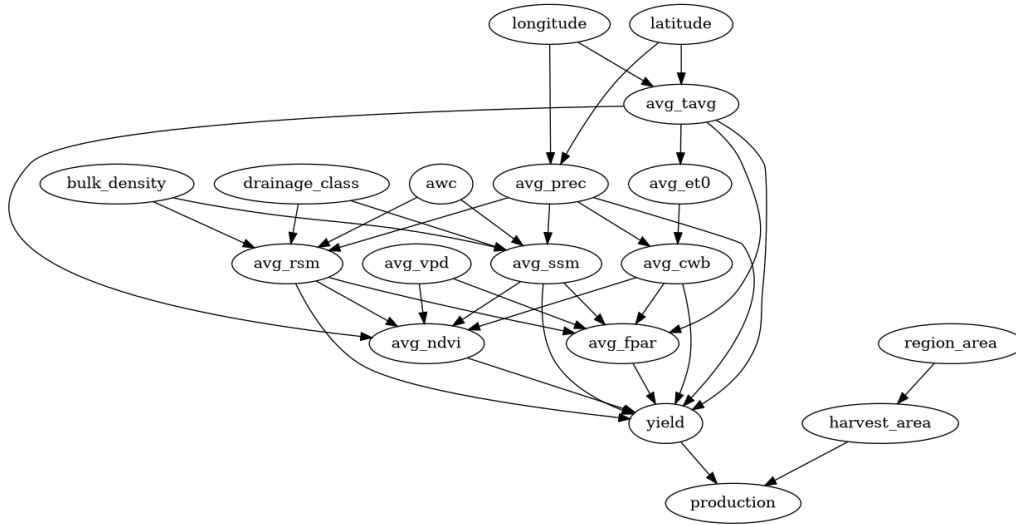


Figure 3.11: Prepared causal DAG for the wheat analysis.

A new domain-driven DAG was constructed that linked soil properties, precipitation, temperature-related variables, soil moisture, vegetation indicators, yield, and production. This DAG was explicitly written as a causal hypothesis rather than being treated as a purely correlation-derived graph.

The resulting structure was then used in a DoWhy workflow. In that stage, the panel was loaded, a treatment-outcome question was specified, an estimation dataset was formed, a causal model was built from the DAG, and the causal effect was identified and

estimated using the ‘backdoor.linear_regression’ estimator, followed by refutation checks using random common cause and placebo treatment tests.

3.6 Predictive Modelling

3.6.1 Traditional Machine Learning Models

After identifying the final causal feature set through the causal inference framework, predictive modelling was performed using four widely adopted machine learning algorithms. The objective was to evaluate whether models trained using only causally relevant variables could achieve prediction accuracy comparable to models trained using the complete feature set.

Two different feature configurations were considered:

- **All Features:** A dataset containing all available numerical predictor variables after removing the target variable (*yield*) and leakage variables such as *production*. This configuration consisted of 23 predictor variables.
- **Causal Features:** A reduced dataset containing only the variables identified through causal analysis, namely:
 - Average Soil Moisture (`avg_ssm`)
 - Average Temperature (`avg_tavg`)
 - Average Solar Radiation (`avg_rad`)
 - Average Climatic Water Balance (`avg_cwb`)
 - Average NDVI (`avg_ndvi`)
 - Average FAPAR (`avg_fpar`)
 - Available Water Capacity (`awc`)
 - Bulk Density (`bulk_density`)
 - Drainage Class (`drainage_class`)

The dataset was divided into training and testing sets using an 80:20 split with a fixed random seed of 42 to ensure reproducibility. The same train-test partition was used for both feature configurations, enabling a fair comparison of model performance.

The following regression algorithms were implemented:

- Linear Regression
- Random Forest Regressor

- XGBoost Regressor
- LightGBM Regressor

For each algorithm, two separate models were trained:

1. Model trained using all predictor variables.
2. Model trained using only the selected causal features.

Model performance was evaluated using the following regression metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

To quantify the effectiveness of causal feature selection, the percentage of predictive performance retained by the causal models was computed as

$$\text{Performance Retained (\%)} = \frac{R^2_{\text{Causal}}}{R^2_{\text{All Features}}} \times 100.$$

This metric measures how much predictive capability is preserved after reducing the feature space using causal reasoning.

The trained models, prediction results, and performance summaries were stored for subsequent comparison and interpretation. Additionally, comparative visualizations were generated to analyse the prediction accuracy of models trained using all features and the reduced causal feature set.

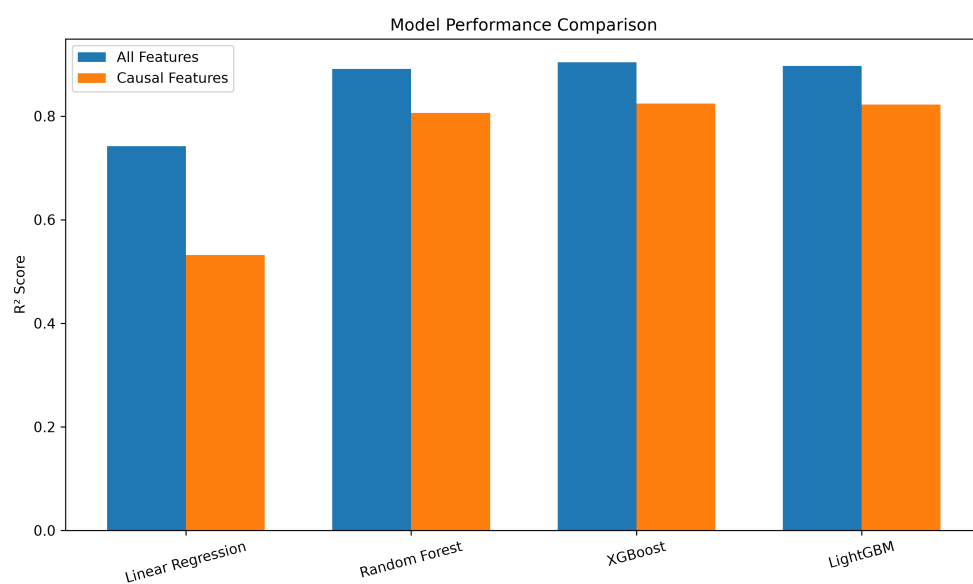


Figure 3.12: Comparison of machine learning models using all features and causally selected features.

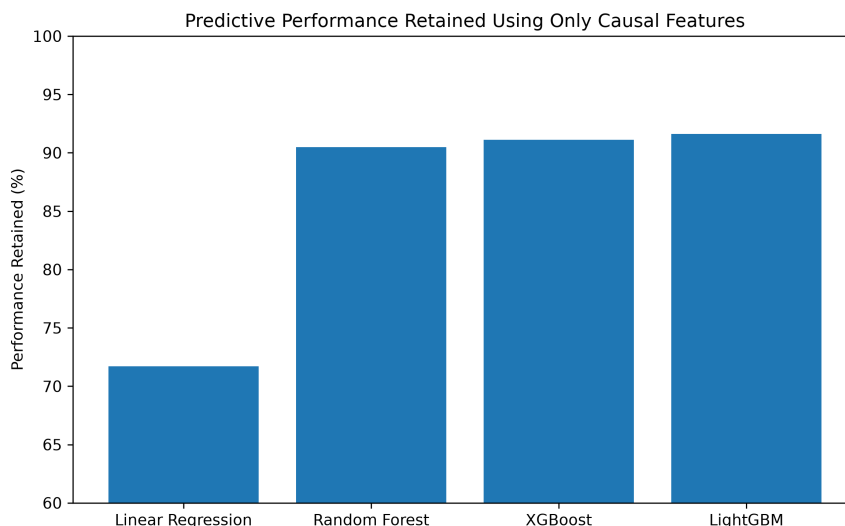


Figure 3.13: Percentage of predictive performance retained after causal feature selection.

3.6.2 Deep Learning Models

In addition to the traditional machine learning algorithms, deep learning models were implemented to investigate their capability in modelling the complex nonlinear relationships between climatic, soil, vegetation, and crop yield variables. Two neural network architectures were considered in this study: a Multi-Layer Perceptron (MLP) and a Long Short-Term Memory (LSTM) network. Both models were trained using (i) the complete predictor set and (ii) the causal feature subset obtained from the causal inference framework. This comparison was performed to evaluate whether causal feature selection could reduce the input dimensionality while maintaining predictive performance.

Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron (MLP) is a feed-forward artificial neural network capable of approximating complex nonlinear functions through multiple hidden layers. Prior to training, all predictor variables were standardised using the `StandardScaler` to ensure that each feature contributed equally during optimisation.

The network architecture consisted of three hidden layers containing 128, 64, and 32 neurons, respectively. The Rectified Linear Unit (ReLU) activation function was employed for all hidden layers, while the Adam optimiser with an initial learning rate of 0.001 was used for weight optimisation. The maximum number of training iterations was set to 500. To reduce overfitting, early stopping was enabled using a validation fraction of 10%, allowing training to terminate automatically when no further improvement in validation performance was observed.

The trained model was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Training loss curves

and actual-versus-predicted plots were also generated to assess convergence and predictive behaviour.

Long Short-Term Memory (LSTM)

To further investigate the applicability of deep learning techniques, a Long Short-Term Memory (LSTM) network was implemented. Although LSTM models are primarily designed for sequential and temporal data, they were included in this study to examine their ability to learn nonlinear feature interactions within the crop yield dataset.

Before training, all input variables were standardised using the `StandardScaler`. Since the dataset consists of annual observations rather than multi-step sequences, each sample was reshaped into a single time step, resulting in an input tensor of dimension (samples, 1, features).

The LSTM architecture comprised a single LSTM layer with 64 hidden units followed by a dropout layer with a dropout rate of 0.20 to minimise overfitting. Two fully connected dense layers containing 32 and 16 neurons with ReLU activation were then employed before the final output neuron used for yield prediction. The network was trained using the Adam optimiser with Mean Squared Error (MSE) as the loss function, while Mean Absolute Error (MAE) was monitored as an additional evaluation metric.

Training was performed for a maximum of 200 epochs with a batch size of 32. Early stopping with a patience of 20 epochs was incorporated to restore the best-performing model based on validation loss and prevent unnecessary training once convergence had been achieved.

Similar to the MLP model, both the full feature set and the causal feature subset were used for model training. Performance was subsequently evaluated using MAE, RMSE, and R^2 , together with learning curves, residual plots, and actual-versus-predicted visualisations.

Implementation Details

Both deep learning models were implemented using Python. The MLP model was developed using the `scikit-learn` library, whereas the LSTM network was implemented using TensorFlow and Keras. Model parameters, trained weights, feature scalers, prediction outputs, and evaluation reports were stored to ensure reproducibility and facilitate subsequent comparison with traditional machine learning models.

3.6.3 Model Validation

To ensure a fair and consistent comparison between all predictive models, a common validation strategy was adopted throughout the study. The processed dataset was randomly

divided into training and testing subsets using an 80:20 split, where 80% of the observations were used for model training and the remaining 20% were reserved for evaluating predictive performance on unseen data. A fixed random seed (`random_state = 42`) was used during the train–test split to ensure reproducibility of the experimental results.

Prior to model training, numerical predictor variables were standardised using the `StandardScaler`. The scaler was fitted only on the training data and subsequently applied to both the training and testing sets to avoid data leakage. Feature scaling was performed for the Multi-Layer Perceptron (MLP) and Long Short-Term Memory (LSTM) models, while tree-based algorithms such as Random Forest, XGBoost, and LightGBM were trained on the original feature values.

To investigate the impact of causal feature selection, every predictive model was trained under two different experimental settings:

- Using the complete set of predictor variables.
- Using only the causal feature subset identified through the causal inference framework.

The predictive performance of each model was evaluated using three widely adopted regression metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

Mean Absolute Error measures the average magnitude of prediction errors, Root Mean Squared Error gives greater emphasis to larger errors, and the coefficient of determination (R^2) quantifies the proportion of variance in crop yield explained by the predictive model.

In addition to these quantitative metrics, several graphical diagnostic techniques were employed to assess model behaviour and predictive quality. These included training loss curves for deep learning models, Actual versus Predicted scatter plots to examine prediction accuracy, and residual plots to identify systematic prediction errors and evaluate the distribution of residuals.

The use of identical training, testing, and evaluation procedures across all machine learning and deep learning models ensured that the comparative analysis reflected differences in modelling capability rather than differences in the experimental setup.

3.7 Model Interpretation and Comparison

To evaluate the effectiveness of the proposed causal inference framework, the predictive performance of all machine learning and deep learning models was systematically compared under two different feature configurations: (i) the complete set of predictor variables and (ii) the causal feature subset identified through the causal analysis.

The comparison focused on determining whether the reduced causal feature set could achieve predictive performance comparable to models trained using the complete feature space. For each algorithm, regression performance was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

In addition to quantitative performance metrics, model interpretability was incorporated into the evaluation process. For tree-based machine learning models, SHapley Additive exPlanations (SHAP) were employed to quantify the contribution of individual predictor variables toward crop yield predictions. SHAP values provide both global and local explanations by estimating the marginal contribution of each feature to the model output based on cooperative game theory.

The generated SHAP summary plots and feature importance visualisations were used to identify the most influential environmental and soil-related variables contributing to yield prediction. Particular emphasis was placed on examining whether the features selected through the causal inference framework also emerged as the most influential variables from the predictive models.

For the deep learning models, interpretability was primarily achieved through prediction diagnostics rather than feature attribution. Training and validation loss curves were analysed to monitor convergence behaviour, while Actual versus Predicted plots and residual plots were used to assess prediction accuracy and identify potential systematic errors.

Finally, the overall comparison considered both predictive performance and model complexity. Models trained using the causal feature subset were evaluated to determine whether they could maintain high predictive accuracy while reducing the dimensionality of the input space. This comparison provides insight into the practical benefits of integrating causal inference with predictive modelling for crop yield forecasting.

3.8 Development of the Decision Support System

To demonstrate the practical applicability of the proposed crop yield forecasting framework, a Decision Support System (DSS) was developed using the Streamlit framework. The application provides an interactive interface through which users can estimate crop yield by supplying relevant climatic, vegetation, and soil characteristics. The primary

objective of the DSS is to translate the developed predictive models into an accessible tool that can support agricultural decision-making.

The application was implemented in Python using the Streamlit library for the graphical user interface. The trained predictive models, feature scalers, and preprocessing pipeline were integrated into the application to ensure consistency between the training environment and real-time prediction. The workflow of the Decision Support System is illustrated in Figure ??.

The operational workflow of the application consists of the following steps:

1. The user enters the required climatic, vegetation, and soil parameters through the graphical interface.
2. The input variables are automatically preprocessed using the same feature scaling procedure applied during model training.
3. The selected predictive model is loaded from the saved model files.
4. The processed input is passed to the predictive model to estimate crop yield.
5. The predicted crop yield is displayed to the user together with the input summary.

To improve usability, the application includes input validation to ensure that only valid numerical values are accepted. Default values and descriptive labels are provided for each predictor variable to assist users during data entry. The interface was designed to remain lightweight and responsive, allowing predictions to be generated within a few seconds.

The deployment architecture separates the user interface from the trained predictive models. All trained models, feature scalers, and required preprocessing objects were stored locally and loaded dynamically when predictions were requested. This modular design allows individual predictive models to be replaced or updated without modifying the application interface.

The developed Decision Support System serves as a proof-of-concept implementation of the proposed crop yield forecasting framework and demonstrates how causal feature selection and predictive modelling can be integrated into an interactive decision-support application suitable for researchers, agricultural practitioners, and policymakers.

Chapter 4 Data Analysis and Results

4.1 Exploratory Data Analysis Results

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the processed crop yield dataset before causal analysis and predictive modelling. The objective was to examine the distribution of crop yield, identify relationships among climatic and soil variables, detect potential outliers, and understand the underlying data structure. The findings obtained during EDA provided important insights for feature selection and motivated the subsequent causal inference analysis.

4.1.1 Distribution of Crop Yield

Figure 4.1 illustrates the distribution of crop yield values in the processed dataset.

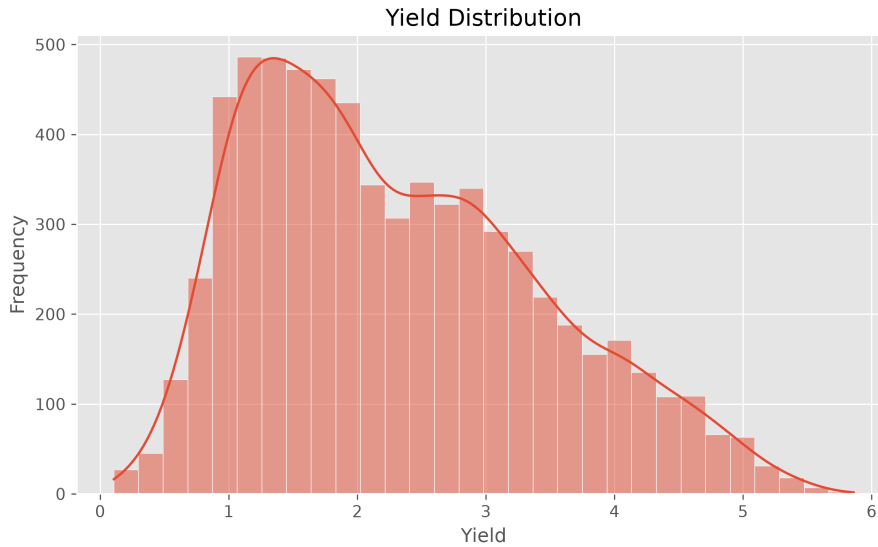


Figure 4.1: Distribution of Crop Yield

The histogram indicates that crop yield values exhibit a moderately skewed distribution with most observations concentrated around the central range. Only a small number of observations correspond to extremely high or low yields, suggesting that the dataset contains relatively few outliers. Such a distribution is suitable for supervised regression

modelling after appropriate feature scaling.

4.1.2 Correlation Analysis

To understand the linear relationship among numerical variables, Pearson's correlation matrix was computed.

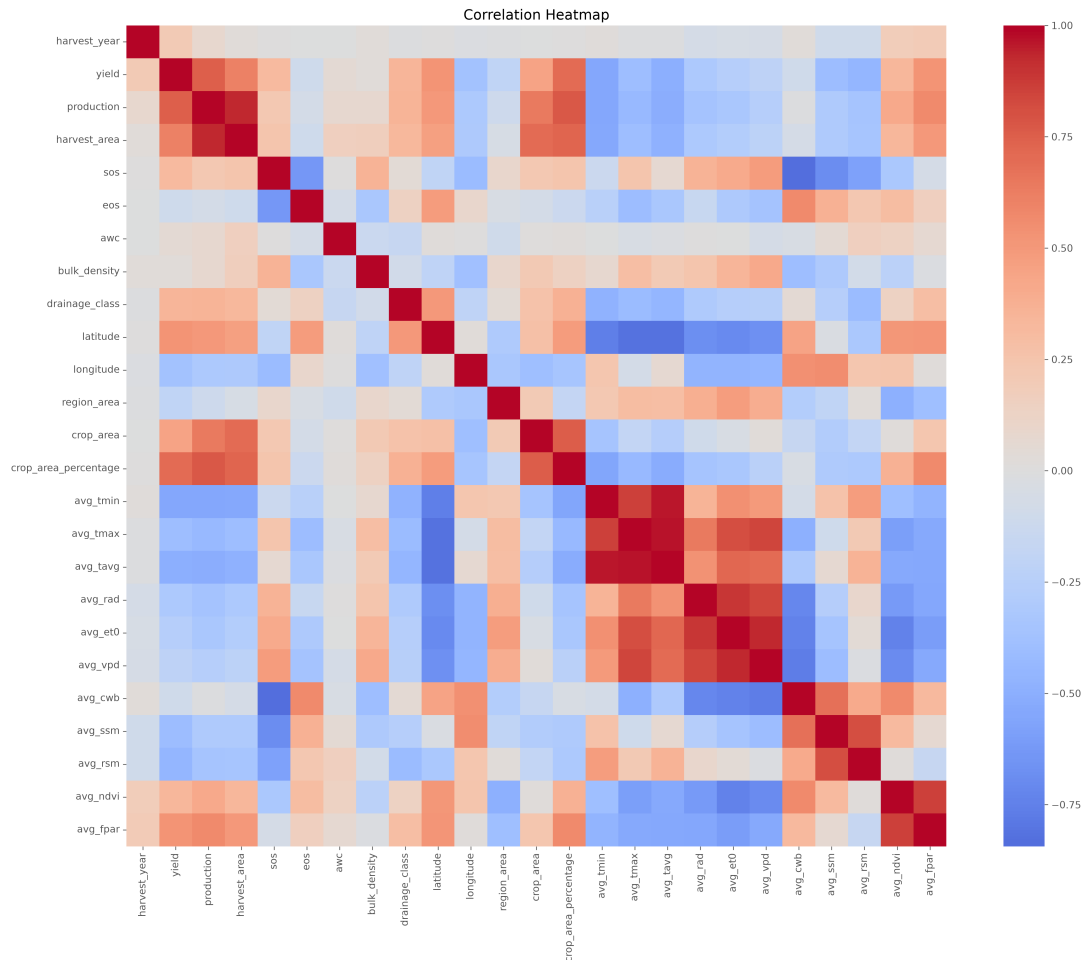


Figure 4.2: Correlation Heatmap of Numerical Features

The correlation heatmap reveals that vegetation indices such as NDVI and FPAR exhibit strong positive correlations with crop yield, whereas climatic variables including average temperature and climatic water balance demonstrate moderate relationships. Soil properties generally show weaker linear correlations, indicating that their influence may be indirect or non-linear. These observations motivated the use of causal inference methods instead of relying solely on correlation analysis.

4.1.3 Feature Relationships

Scatter plots and pairwise visualizations were generated to examine the relationship between crop yield and important climatic, vegetation, and soil variables.

The visualizations indicate that vegetation-related features exhibit clearer positive trends with crop yield compared to several climatic and soil variables. However, many relationships are non-linear, suggesting that conventional linear statistical measures alone are insufficient to capture the underlying dependencies. This observation further justifies the use of machine learning and causal inference techniques for modelling crop yield.

4.1.4 Summary of EDA Findings

The exploratory analysis provided the following key observations:

- Crop yield follows a moderately skewed distribution with limited extreme observations.
- Vegetation indices (NDVI and FPAR) exhibit stronger positive relationships with crop yield than most climatic and soil variables.
- Climatic variables such as average temperature, radiation, and climatic water balance show moderate associations with yield.
- Soil characteristics demonstrate relatively weaker linear relationships but may contribute indirectly through complex interactions.
- Correlation analysis alone cannot distinguish causal relationships from spurious associations. Therefore, a causal inference framework was adopted to identify variables with genuine causal influence on crop yield.

4.2 Causal Analysis Results

The primary objective of the causal analysis was to distinguish genuine cause-and-effect relationships from simple statistical correlations. Unlike traditional feature selection techniques that rely only on predictive importance, causal inference identifies variables that directly influence crop yield while reducing the effect of confounding factors. This study combines expert agricultural knowledge with data-driven causal discovery to obtain a robust causal graph before estimating treatment effects and selecting causal predictors.

4.2.1 Expert Agricultural Directed Acyclic Graph (DAG)

An expert-driven Directed Acyclic Graph (DAG) was constructed using established agricultural knowledge and previous literature. The graph represents hypothesized causal relationships among climatic variables, soil properties, vegetation indices, and crop yield.

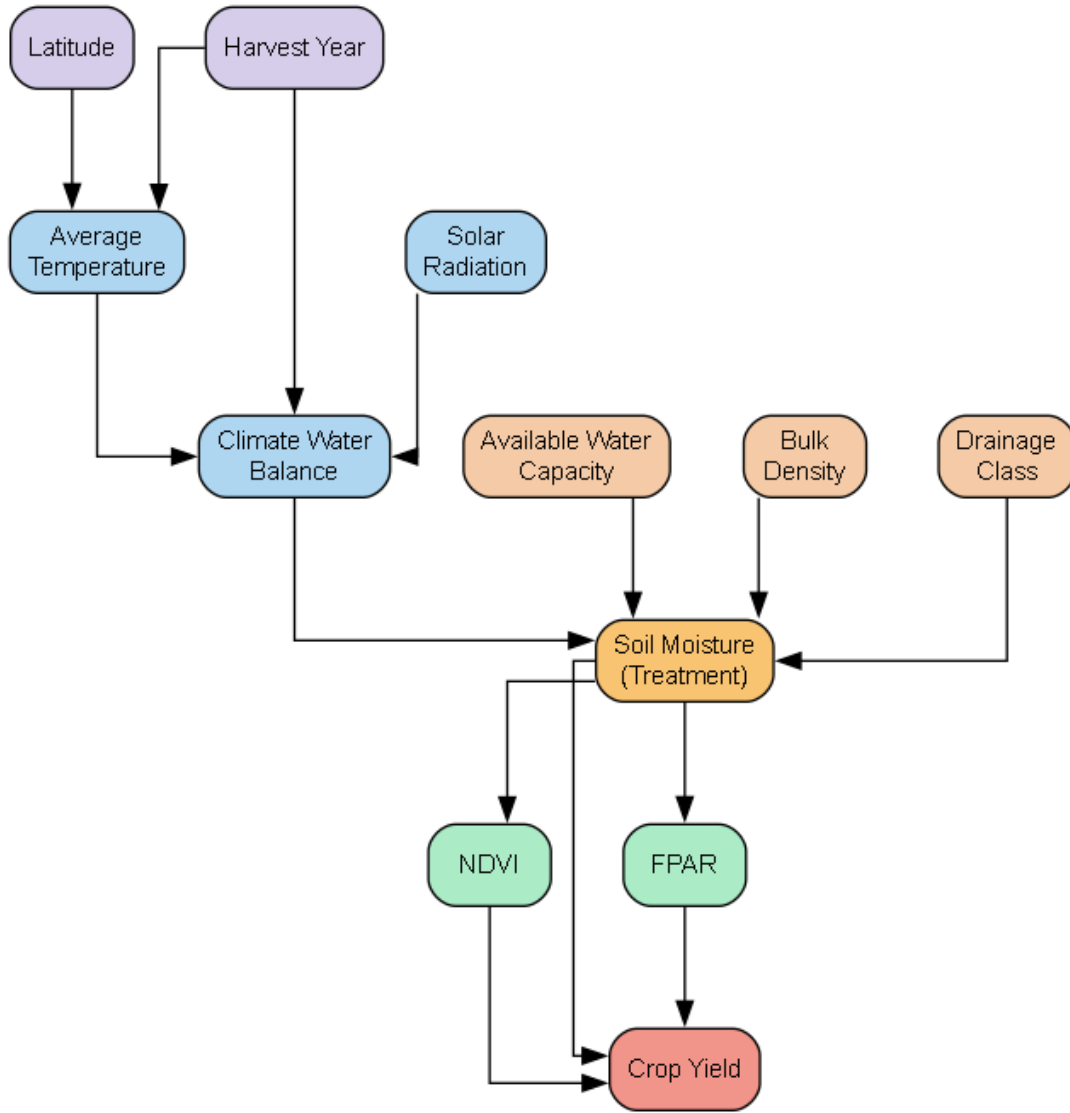


Figure 4.3: Expert Agricultural Directed Acyclic Graph

The expert DAG assumes that climatic variables directly influence vegetation growth and soil moisture, which subsequently affect crop yield. Soil characteristics such as available water capacity, bulk density, and drainage class are considered relatively stable factors that influence water availability and root development. Vegetation indices including NDVI and FPAR act as intermediate variables representing crop health and biomass accumulation.

4.2.2 Data-driven Causal Discovery using NOTEARS

To validate and complement the expert knowledge, a data-driven causal discovery algorithm based on NOTEARS (Non-combinatorial Optimization via Trace Exponential and Augmented Lagrangian for Structure Learning) was applied to the processed dataset.

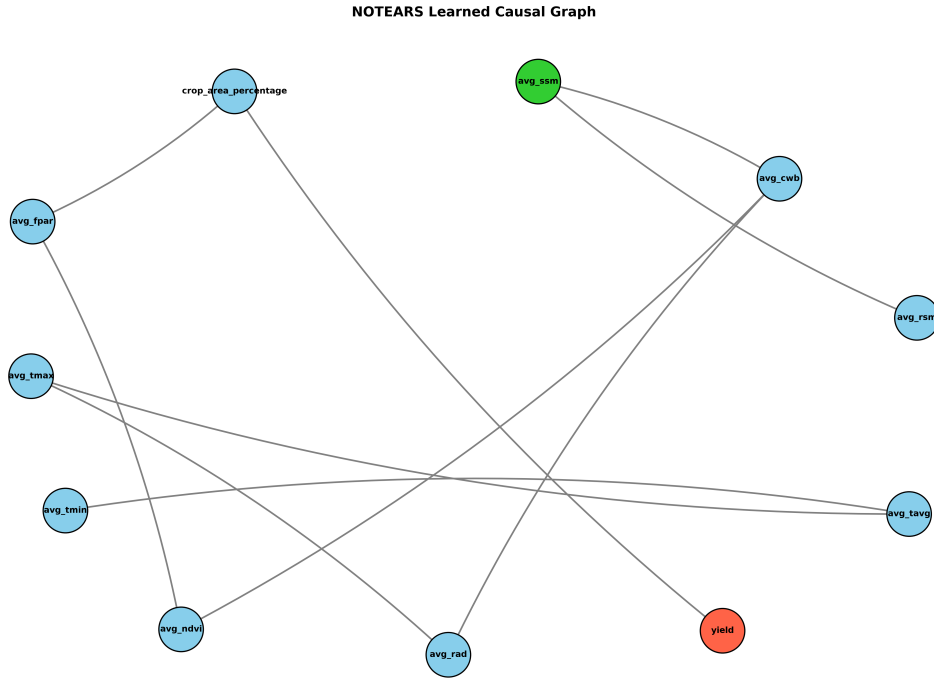


Figure 4.4: Data-driven Causal Graph Generated using NOTEARS

The NOTEARS algorithm learns the underlying causal structure directly from observational data while enforcing the acyclic property of the graph. Several causal relationships identified by NOTEARS were consistent with expert agricultural knowledge, particularly the influence of vegetation indices and climatic variables on crop yield. Minor structural differences were also observed, highlighting potential relationships that warrant further agricultural investigation.

4.2.3 Treatment and Outcome Selection

Based on the causal graphs and domain knowledge, crop yield was selected as the outcome variable. Climatic, vegetation, and soil-related variables that demonstrated direct causal influence were considered as treatment variables for subsequent causal effect estimation.

The selected treatment variables included:

- Average Soil Moisture (avg_ssm)
- Average Temperature (avg_tavg)
- Average Radiation (avg_rad)
- Average Climatic Water Balance (avg_cwb)
- Average NDVI (avg_ndvi)
- Average FPAR (avg_fpar)

- Available Water Capacity (awc)
- Bulk Density
- Drainage Class

These variables formed the causal feature subset that was later used for predictive modelling.

4.2.4 Causal Effect Estimation

The Average Treatment Effect (ATE) was estimated using stabilized Inverse Probability Weighting (IPW). Stabilized weights were adopted to reduce variance and improve estimation stability.

The estimated treatment effect obtained from the stabilized IPW model is summarized in Table 4.1.

Table 4.1: Average Treatment Effect Estimation

Metric	Value
Estimated ATE	-0.001715
True ATE	-0.000442
Bias	-0.001273
Propensity Score ROC-AUC	0.725

The negative estimated ATE suggests that the selected treatment variable produced a slight reduction in the expected crop yield under the specified intervention. Furthermore, the relatively small estimation bias indicates that the stabilized weighting approach successfully reduced confounding effects while producing consistent treatment effect estimates.

4.2.5 Causal Feature Selection

The final causal feature subset was obtained by integrating expert agricultural knowledge, data-driven causal discovery, and causal effect estimation. From the original set of numerical predictors, only nine variables were identified as having meaningful causal influence on crop yield.

The selected causal predictors are listed in Table 4.2.

Table 4.2: Final Selected Causal Features

Feature Category	Selected Variables
Climate	avg_ssm, avg_tavg, avg_rad, avg_cwb
Vegetation	avg_ndvi, avg_fpar
Soil	awc, bulk_density, drainage_class

The selected variables were subsequently used to train both traditional machine learning models and deep learning models. Experimental results demonstrate that the reduced causal feature subset retained nearly the same predictive performance as the complete feature set while improving interpretability and reducing model complexity. This confirms that causal feature selection can effectively eliminate redundant predictors without substantial loss in forecasting accuracy.

4.2.6 Causal Discovery using CCM and Analysis using DoWhy

First CCM Run

Convergent Cross Mapping (CCM) was applied to selected variable pairs to study nonlinear relationships in the prepared panel. The CCM skill measures how well the state of one variable can be reconstructed from the time-series history of another variable. A higher CCM skill indicates stronger shared dynamical structure between the two variables.

The chosen CCM pairs are as follows:

- cwb_ssm
- ssm_fpar
- ssm_ndvi
- ssm_yield
- tavg_cwb

For the results discussion, the report focuses on the strongest and most interpretable pairs, especially ssm_yield, ssm_ndvi, and ssm_fpar.

The CCM pairs were selected from the expert DAG, so the analysis focused on soil moisture, climate, vegetation, and yield relationships that were already plausible from the causal structure.

The first set of results was examined through CCM convergence plots, where the cross-map skill is shown as a function of library size. "Library size" refers to the amount of data used for reconstructing the underlying state space of the system. If the CCM skill increases and then stabilizes as library size increases, this is usually taken as evidence that the two variables are dynamically related rather than being linked only by noise.

Among the tested pairs, the relationship between soil surface moisture and yield ('ssm_yield') showed a gradual increase in CCM skill with increasing library size, followed by a tendency toward stabilization. This suggests that the two variables are not independent and may share an underlying nonlinear dynamical relationship. The same general pattern was also seen for pairs such as 'ssm_ndvi' and 'ssm_fpar', while the 'cwb_ssm' pair showed even stronger convergence and therefore appeared to be one of the strongest coupled relationships in the first run.

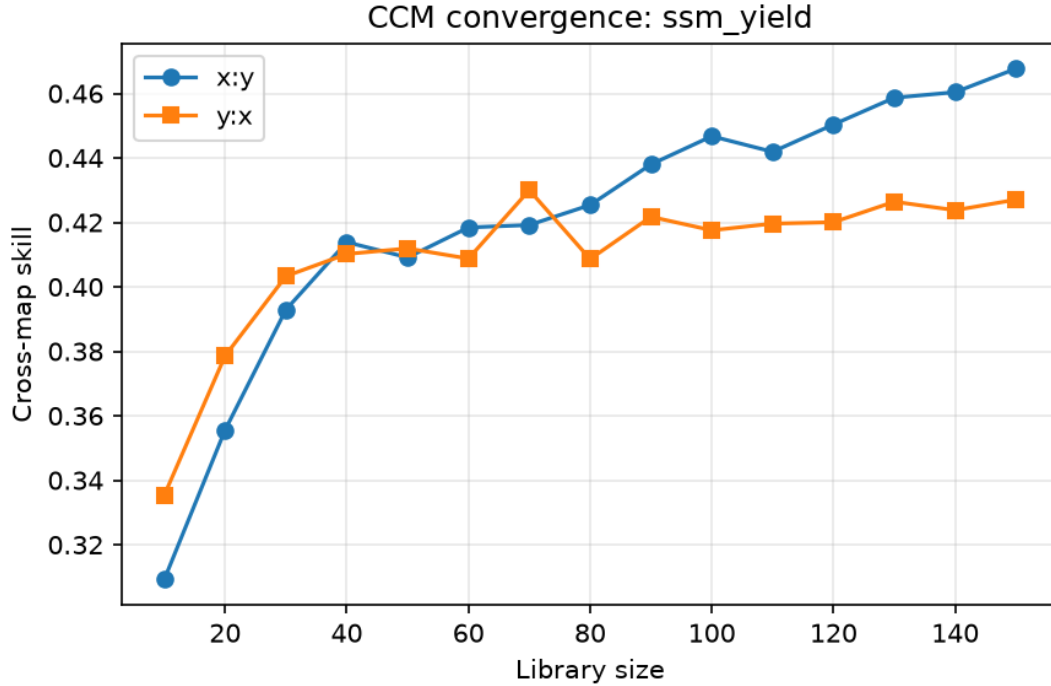


Figure 4.5: CCM convergence for the ssm_yield pair. The cross-map skill increases with library size and begins to stabilize at larger library sizes.

A second diagnostic was carried out using null comparisons. In these plots, the observed CCM skill is compared with the distribution obtained from randomized or shuffled data. For the 'ssm_yield', 'ssm_ndvi', and 'ssm_fpar' pairs, the observed CCM skill lay far above the null distribution. This indicates that the measured cross-map performance is unlikely to be explained by random chance alone.

A third diagnostic considered the role of the embedding dimension E . In CCM, the embedding dimension specifies how many lagged coordinates are used to reconstruct the system dynamics in state space. The sensitivity plots for 'ssm_yield', 'ssm_ndvi', and 'ssm_fpar' showed that the final CCM skill generally increased as E increased, and then began to level off. This suggests that the results are reasonably stable under changes in reconstruction dimension and are not produced by a single arbitrary parameter choice.

Overall, the first CCM run suggests that the chosen pairs are dynamically connected and that the relationships are not random. The strongest support came from ssm_yield, while ssm_ndvi and ssm_fpar also showed clear non-random coupling. However, the

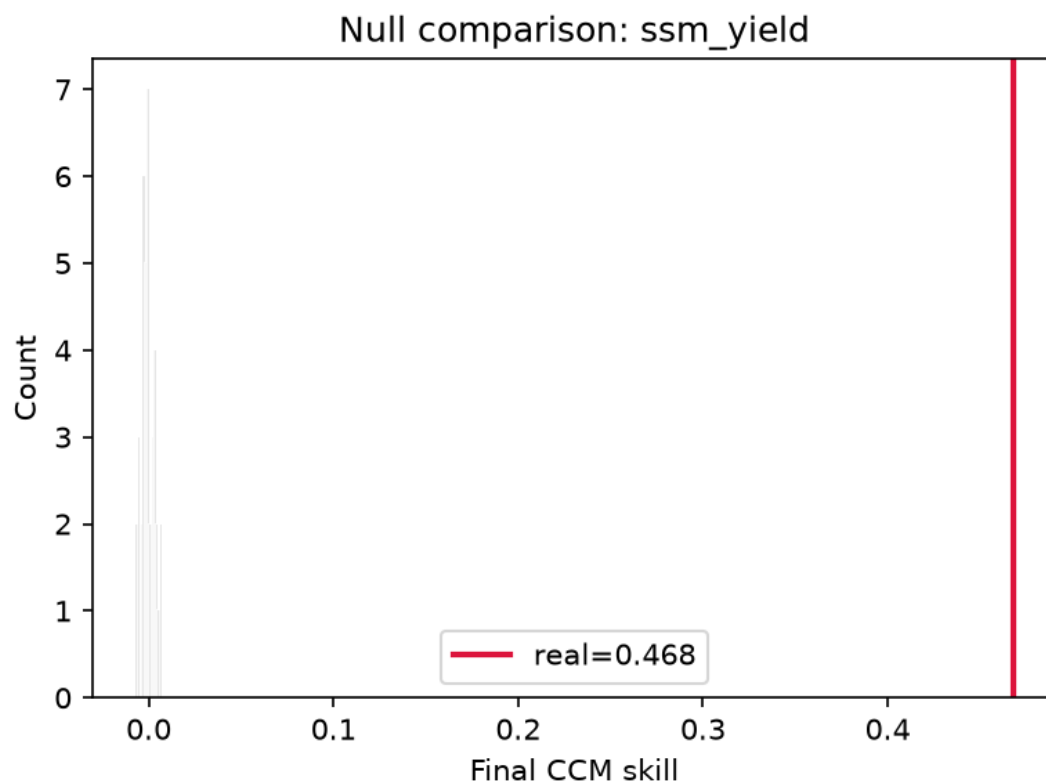


Figure 4.6: Null comparison for the ssm_yield pair. The observed final CCM skill lies far above the null distribution obtained from randomized data.

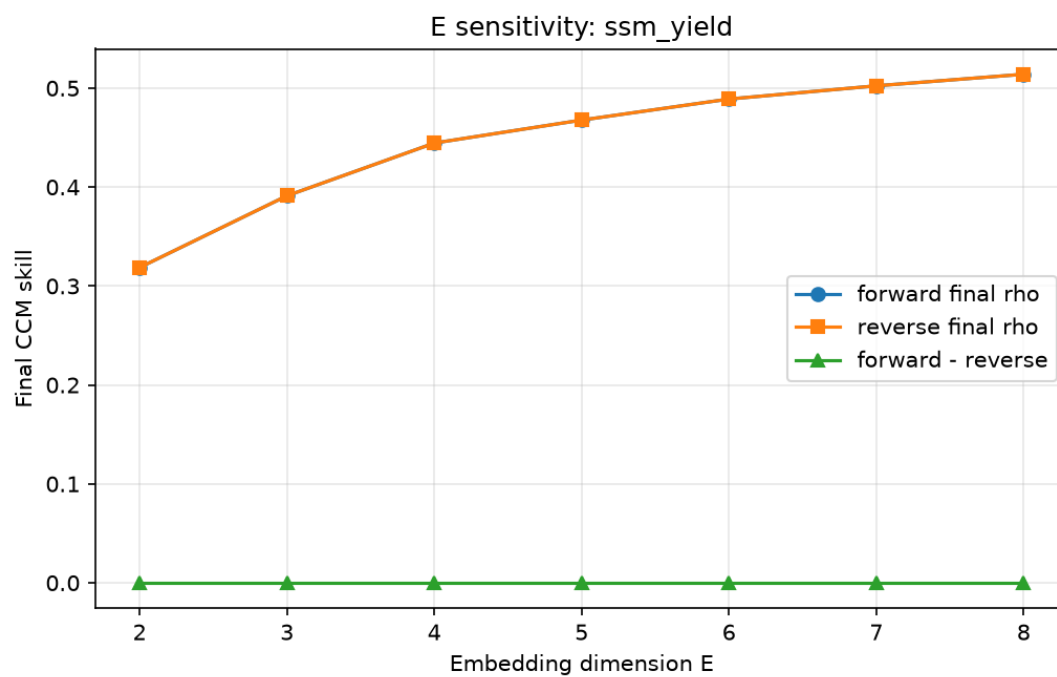


Figure 4.7: Embedding-dimension sensitivity for the ssm_yield pair. The final CCM skill increases with embedding dimension and then levels off.

CCM summary does not support a strong one-way causal claim on its own, so the result should be described as evidence of nonlinear dependence and shared dynamical structure.

For readability, only the most representative figures are included here; the complete set of CCM plots is available in the project repository.

Pre-processing for second CCM run

After pre-processing, the DAG notebook loaded a panel with 7,552 observations and 45 columns, and then extracted a 30-variable subset for causal-structure work. A missingness check showed that most variables were complete, while ‘avg_ssm’ and ‘avg_rsm’ had some missing values, with missing fractions of about 0.110832.

Second CCM Run

The value of E was kept fixed at 2 to ensure consistency across all CCM runs and because tuning E separately for every pair and every unit would have made the analysis much more expensive computationally. Similarly, library sizes were restricted to small values because the batch CCM computation across hundreds of units was already time-consuming, so a compact library schedule was used as a practical compromise between computational feasibility and exploratory analysis.

The heatmap shows the final CCM mean ρ values for each pair of variables and direction. While most relationships are weak or zero, a few moisture-vegetation-climate links show modest positive signal. The figure is best interpreted as an exploratory screen for directional dependence rather than definitive evidence of causality.

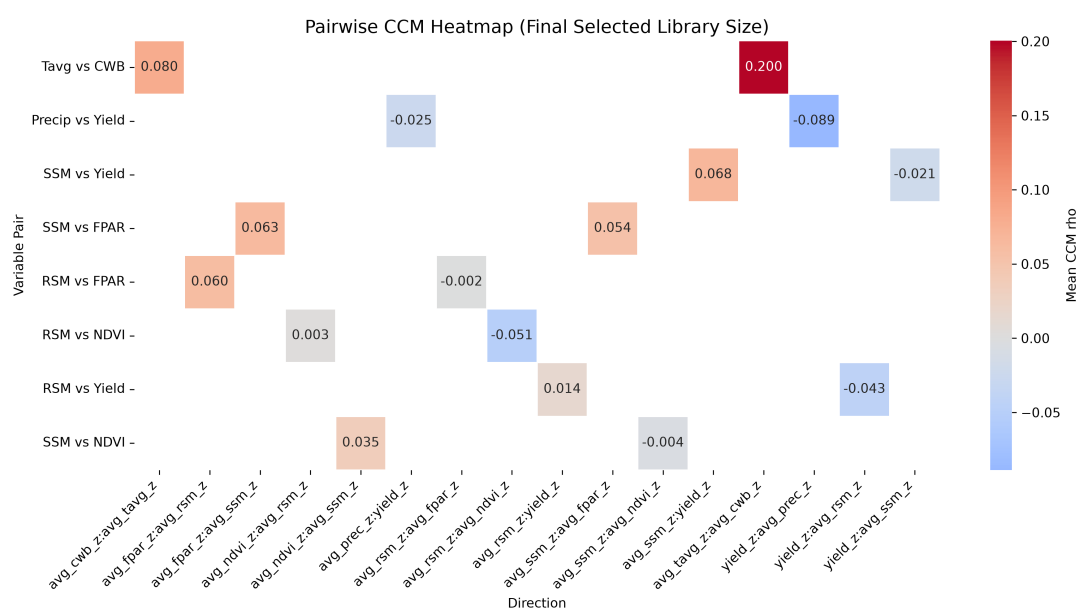


Figure 4.8: Heatmap of final pairwise CCM results across selected variables and directions, with cell color representing the mean ρ at the final library size

DoWhy

After the exploratory CCM analysis, the next step was formal causal estimation using DoWhy. This allowed the candidate relationships to be tested under an explicit adjustment set derived from the DAG, producing estimated effects for soil moisture, vegetation indices, and related variables on yield. The DoWhy results complement the CCM heatmap by converting directional dependence into an interpretable effect-size framework.

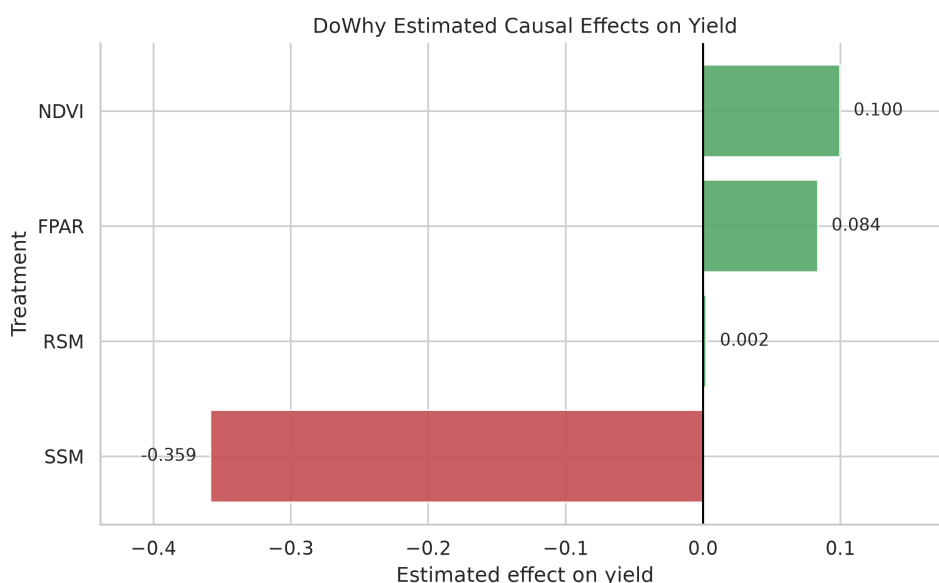


Figure 4.9: Estimated causal effects of selected treatment variables on yield using the DoWhy framework. The plot summarizes the direction and relative magnitude of the estimated effects obtained under the DAG-based adjustment sets.

The estimated effects show substantial variation across variables. The largest magnitude effect is observed for ‘avg_ssm’, which has a negative estimated effect on ‘yield’, whereas ‘avg_ndvi’ and ‘avg_fpar’ have positive estimated effects and ‘avg_rsm’ remains close to zero. This indicates that the contribution of moisture and vegetation-related variables to yield is heterogeneous and likely mediated through multiple interacting pathways.

Table 4.3: DoWhy effect estimates and refutation tests for selected treatment variables.

Treatment	Estimate	Random common cause	Placebo	Subset
avg_ssm	-0.3585	0.98	0.90	0.84
avg_rsm	0.0023	0.76	0.98	0.92
avg_fpar	0.0836	0.98	0.98	1.00
avg_ndvi	0.0996	0.92	0.96	0.84

The refutation tests were used to assess whether the estimated causal effects were sensitive to simple perturbations of the model. Across all four treatments, the random common cause, placebo, and subset refutations remain close to estimates and do not indicate major instability. This suggests that the DoWhy results are reasonably robust

under chosen specification, although they still depend on the validity of the DAG and observed adjustment set.

4.3 Predictive Modelling Results

The predictive performance of both traditional machine learning and deep learning models was evaluated using two feature configurations: (i) the complete feature set and (ii) the causal feature subset obtained through causal inference. The objective was to examine whether the reduced causal feature set could maintain comparable predictive accuracy while improving model interpretability and reducing model complexity.

Table 4.4 summarizes the predictive performance of all evaluated models. The performance retained percentage represents the proportion of predictive accuracy preserved after replacing the complete feature set with the selected causal features.

Table 4.4: Predictive Model Performance Comparison

Model	R^2 (All)	R^2 (Causal)	Retained (%)
Linear Regression	0.742	0.532	71.70
Random Forest	0.891	0.806	90.46
XGBoost	0.904	0.824	91.15
LightGBM	0.897	0.822	91.64
MLP	0.871	0.798	91.62
LSTM	0.874	0.792	90.62

Among the traditional machine learning algorithms, XGBoost achieved the highest predictive performance using the complete feature set with an R^2 score of 0.904, followed closely by LightGBM (0.897) and Random Forest (0.891). Linear Regression recorded the lowest performance, indicating that the relationship between crop yield and the explanatory variables is highly non-linear.

The deep learning models also demonstrated competitive predictive capability. The Multi-Layer Perceptron (MLP) achieved an R^2 score of 0.871, while the Long Short-Term Memory (LSTM) network achieved an R^2 score of 0.874 using the complete feature set. Although these models successfully captured complex non-linear relationships, their performance remained slightly below that of the gradient boosting algorithms.

An important observation is that the predictive accuracy remained remarkably stable after replacing the complete feature set with only the selected causal variables. Random Forest, XGBoost, LightGBM, MLP, and LSTM all retained approximately 90–92% of their original predictive performance despite using fewer input variables. In contrast, Linear Regression retained only 71.70% of its predictive capability, suggesting that linear models are more sensitive to feature reduction than non-linear ensemble and neural network models.

These results demonstrate that the proposed causal feature selection framework effectively removes redundant predictors while preserving most of the predictive information required for accurate crop yield forecasting. Consequently, the causal feature subset

provides a simpler and more interpretable modelling framework without substantially compromising predictive accuracy.

Figure 4.10 presents the comparison of R^2 scores obtained using the complete feature set and the causal feature subset across all evaluated models.

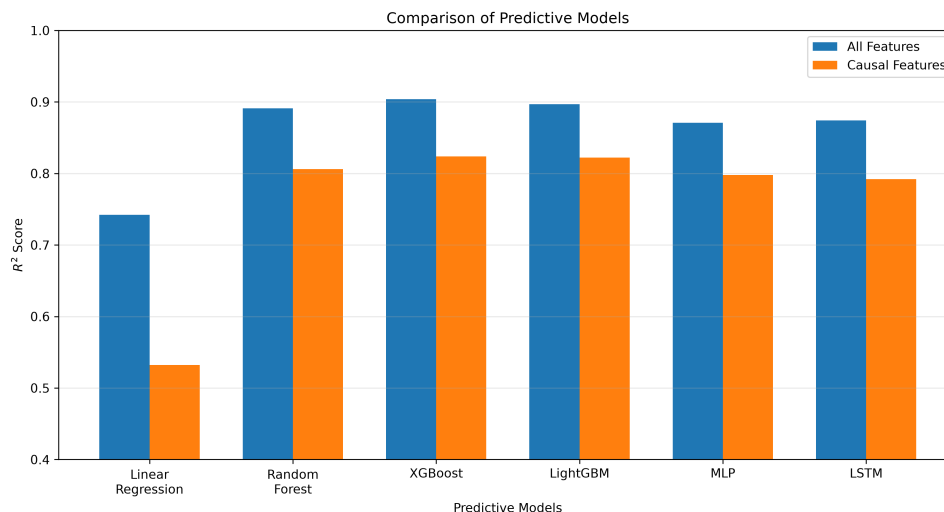


Figure 4.10: Comparison of Predictive Performance Across Machine Learning and Deep Learning Models

4.4 Comparative Performance Analysis

The comparative evaluation of the six predictive models highlights the effectiveness of both ensemble learning techniques and deep learning architectures for crop yield forecasting. While all models benefited from the rich climatic, vegetation, and soil information available in the dataset, clear differences were observed in their predictive capabilities.

Among all evaluated models, XGBoost achieved the highest predictive performance with an R^2 score of 0.904 using the complete feature set. LightGBM and Random Forest closely followed with R^2 scores of 0.897 and 0.891, respectively. These results indicate that ensemble tree-based algorithms are highly effective in capturing the complex non-linear relationships that exist between environmental variables and crop yield.

The deep learning models also demonstrated strong predictive performance. The Long Short-Term Memory (LSTM) network achieved an R^2 score of 0.874, while the Multi-Layer Perceptron (MLP) achieved an R^2 score of 0.871. Although both neural network models successfully modelled non-linear interactions, they did not outperform the gradient boosting algorithms on the structured tabular dataset used in this study.

A notable finding of this research is the minimal reduction in predictive accuracy after replacing the complete feature set with the selected causal feature subset. Except for Linear Regression, all machine learning and deep learning models retained approximately 90%–92% of their original predictive performance. This demonstrates that the

proposed causal feature selection framework successfully eliminated redundant variables while preserving the information most relevant for crop yield prediction.

Linear Regression experienced the largest decline in performance, retaining only 71.70% of its original predictive capability. This suggests that simple linear models are more sensitive to feature reduction and are less capable of representing the complex interactions present in agricultural data compared to non-linear ensemble and neural network approaches.

Overall, the experimental results indicate that causal feature selection offers an effective balance between predictive performance and model simplicity. The reduced feature subset significantly improves interpretability while maintaining competitive forecasting accuracy, thereby supporting the practical deployment of causal machine learning models for agricultural decision support.

4.5 Decision Support System Results

To improve the practical usability of the proposed crop yield forecasting framework, a web-based Decision Support System (DSS) was developed using the Streamlit framework. The application integrates the trained predictive models into an interactive interface, enabling users to estimate crop yield by providing relevant climatic, vegetation, and soil-related parameters.

The objective of the DSS is to bridge the gap between advanced causal machine learning models and real-world agricultural applications. Rather than requiring technical knowledge of machine learning algorithms, users can simply provide the required input variables and obtain an instant prediction of crop yield.

4.5.1 System Interface

The developed Decision Support System provides a simple and user-friendly interface that allows users to enter field-specific, climatic, and vegetation-related information required for crop yield prediction. The interface was developed using Streamlit and is designed to make the prediction process intuitive for users with minimal technical knowledge.

**Causal Crop Advisor**
Helping Farmers Focus on the Factors that Truly Influence Crop Yield

 **Field Information**

Crop	Average Soil Moisture
Wheat	0.32
Harvest Year	Average Temperature (°C)
2020	24.50
Harvest Area (ha)	Solar Radiation
100.00	18.50
Available Water Capacity	Climatic Water Balance
0.18	-8.00
Bulk Density	Average NDVI
1.30	0.45
Drainage Class	Average FPAR
1	0.55

Run Causal Analysis

Figure 4.11: Home page of the Crop Yield Forecasting Decision Support System.

4.5.2 Prediction Module

After entering the required agricultural and environmental variables, the system performs causal analysis and generates crop yield predictions using the trained machine learning models. The prediction module not only estimates crop yield but also provides interpretable recommendations based on the causally important variables identified during the analysis.

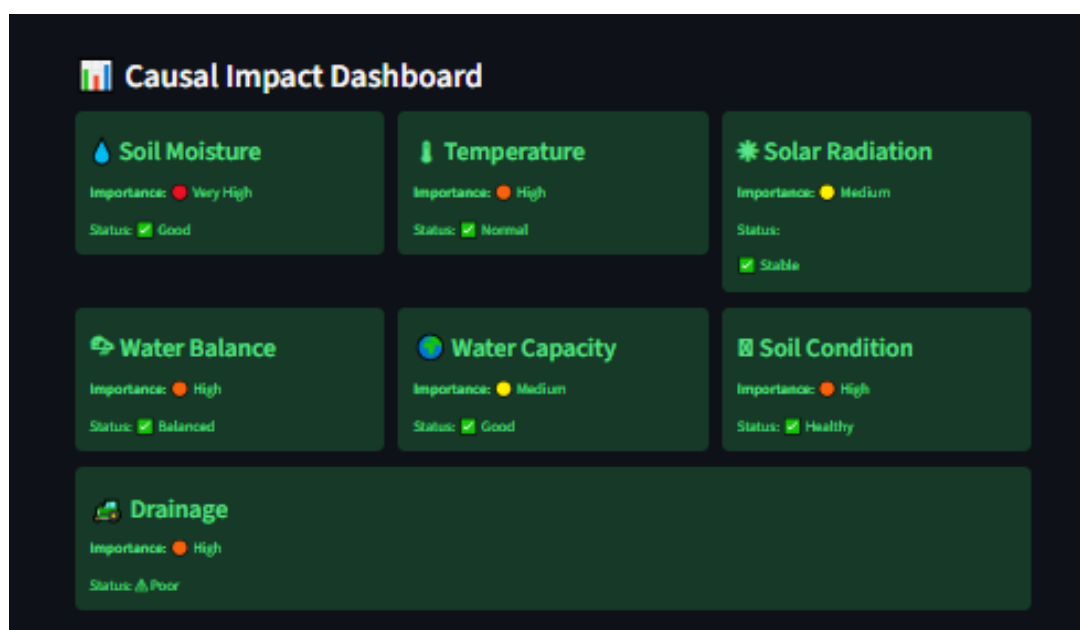


Figure 4.12: Causal Impact Dashboard showing the importance and status of the identified causal variables.

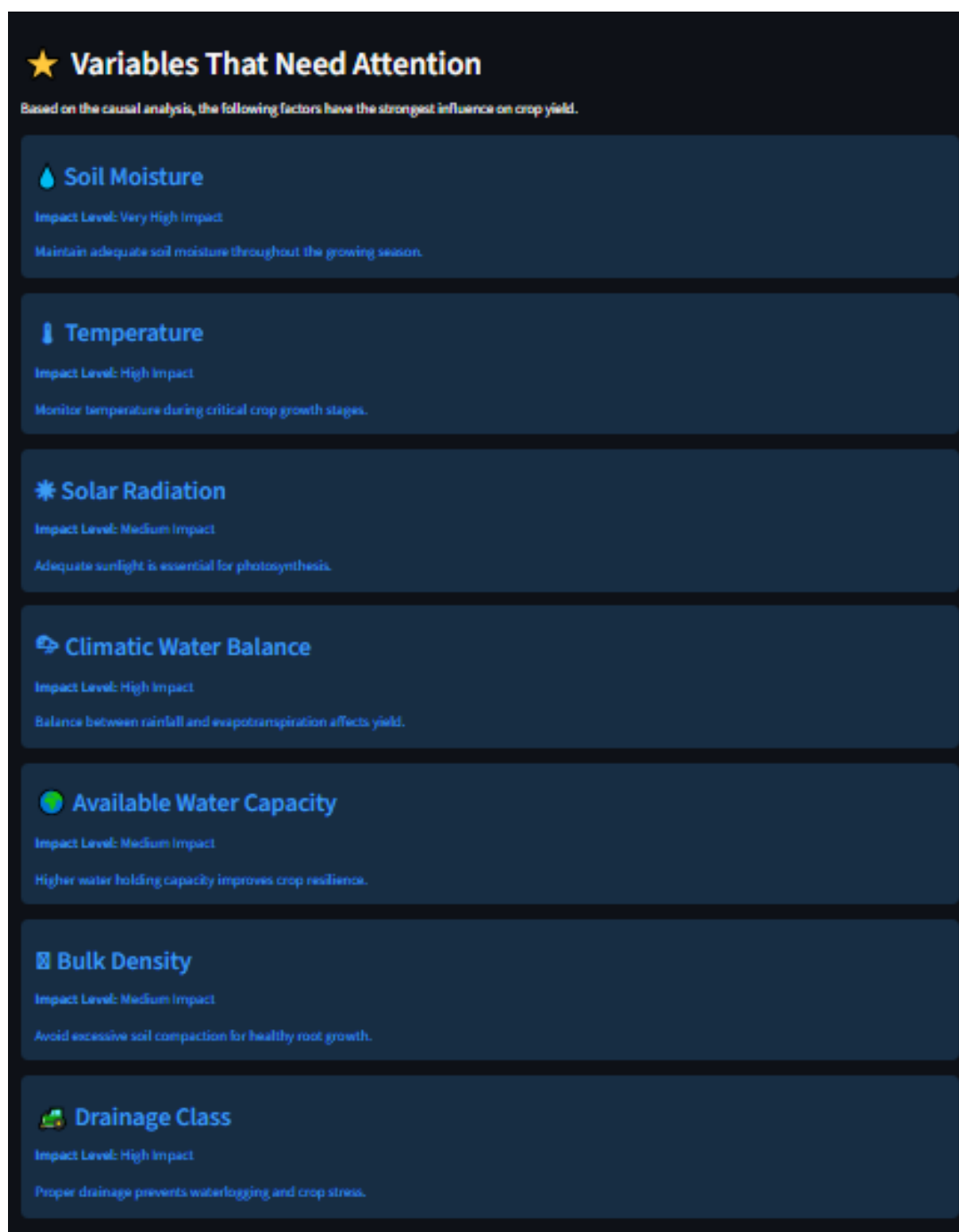


Figure 4.13: Variables requiring attention based on the causal analysis.

4.5.3 Practical Application

The proposed Decision Support System demonstrates the practical applicability of the developed causal machine learning framework. Using only the causally relevant variables identified through the causal inference pipeline, the system predicts crop yield while simultaneously providing explanations and actionable recommendations for farmers and agricultural decision-makers.

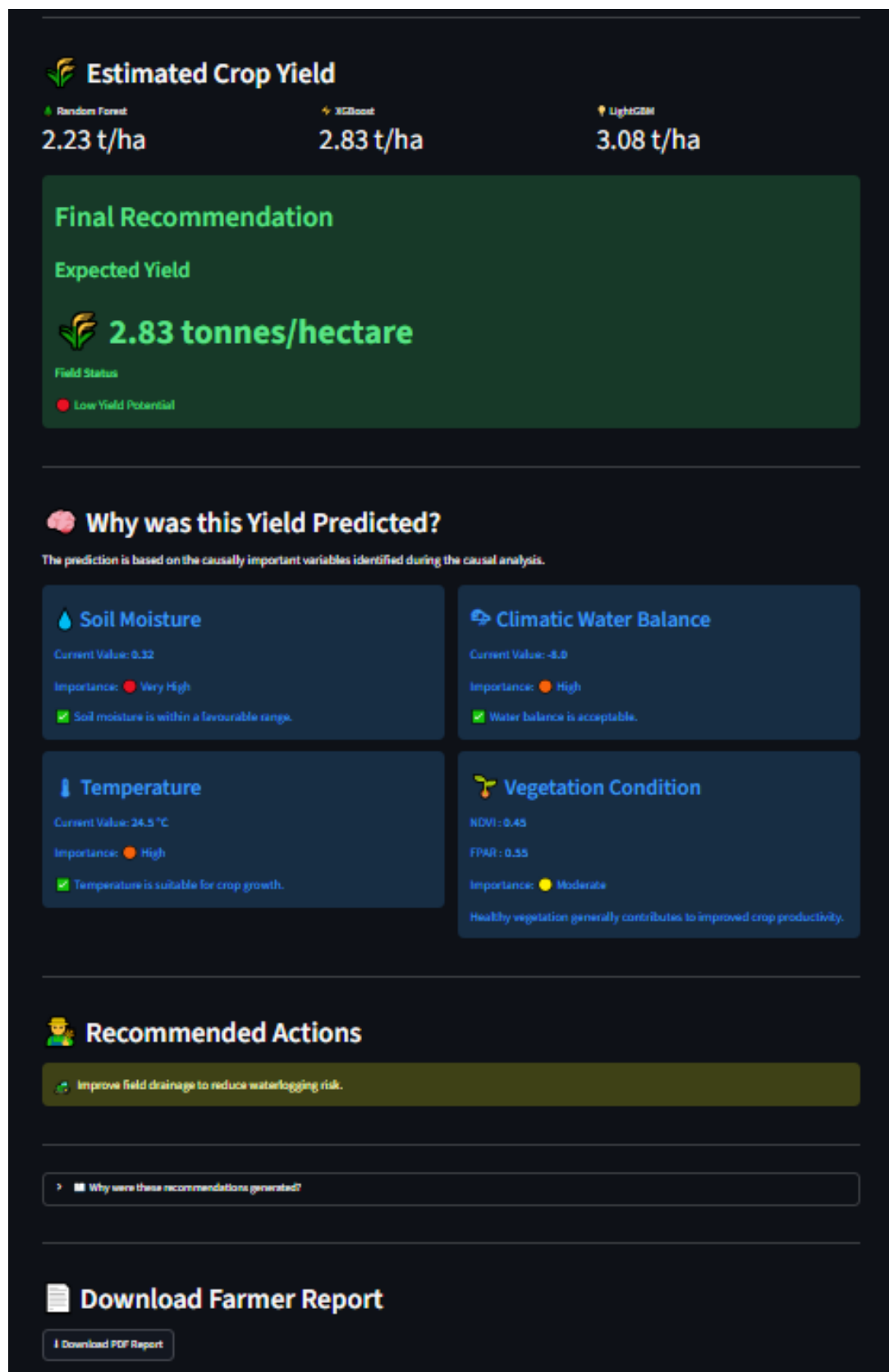


Figure 4.14: Crop yield prediction, model comparison, explanation of predictions, and recommended actions generated by the Decision Support System.

The final interface presents predicted crop yield values generated by multiple predictive models, highlights the most influential causal variables contributing to the prediction, and recommends appropriate agricultural actions. The integrated visualization improves model transparency and enables users to understand not only the predicted yield but also the underlying reasons behind the prediction, making the framework suitable for practical agricultural decision support.

4.5.4 Downloadable Farmer Report

To improve the practical usability of the proposed Decision Support System, a downloadable farmer report was integrated into the application. After the crop yield prediction is generated, the system automatically creates a concise PDF report containing the prediction results, important causal variables, and recommended agricultural actions. This allows users to save the analysis for future reference or share it with agricultural experts and extension officers.

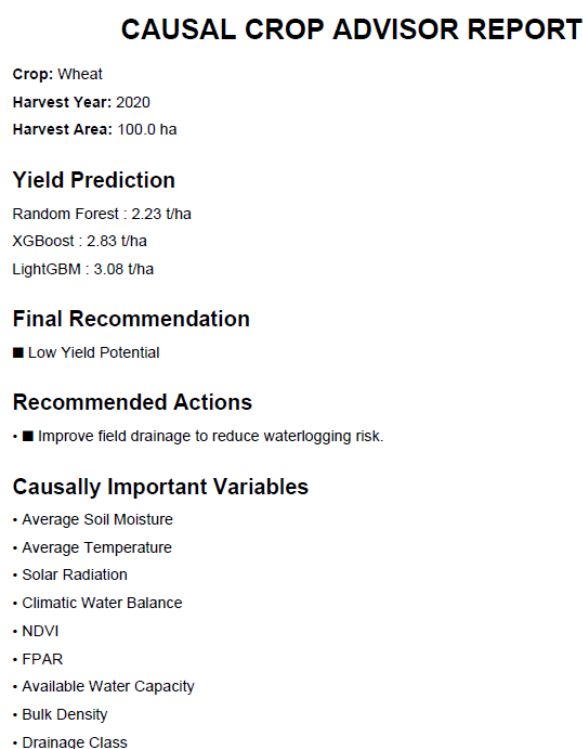


Figure 4.15: Sample PDF report automatically generated by the proposed Crop Yield Forecasting Decision Support System.

The generated report summarizes the crop information provided by the user, predicted yield values obtained from the trained machine learning models, the final recommendation based on the best-performing model, the causally important variables influencing

the prediction, and actionable recommendations for improving crop productivity. The report provides an easy-to-understand summary of the analysis, making the developed framework suitable for real-world agricultural decision support.

Chapter 5 Conclusion

This study presented a causal inference driven framework for crop yield forecasting using the CY-Bench India dataset. Unlike traditional prediction approaches that rely solely on correlation, the proposed methodology integrated causal reasoning with predictive modelling to identify agricultural variables that have a direct influence on crop yield.

A comprehensive data preprocessing and exploratory analysis pipeline was developed to prepare the dataset for causal analysis. Expert agricultural knowledge was first represented using a Directed Acyclic Graph (DAG), followed by data-driven causal discovery using the NOTEARS algorithm. The identified causal relationships were further validated through causal effect estimation using propensity score matching, inverse probability weighting, and doubly robust estimation. Based on these analyses, a subset of causal variables was selected for predictive modelling.

Multiple traditional machine learning models, including Linear Regression, Random Forest, XGBoost, and LightGBM, together with deep learning models such as Multi-Layer Perceptron (MLP) and Long Short-Term Memory (LSTM), were evaluated using both the complete feature set and the causal feature subset. Experimental results demonstrated that the causal feature subset retained more than 90% of the predictive performance for almost all advanced models while using substantially fewer variables. This indicates that the selected causal variables successfully captured most of the information required for accurate crop yield prediction.

Finally, an interactive Streamlit-based Decision Support System was developed to enable users to generate crop yield predictions using the trained models through an intuitive graphical interface. The application demonstrates the practical applicability of the proposed framework for agricultural decision support.

Overall, the study shows that integrating causal inference with machine learning provides an interpretable and computationally efficient alternative to purely correlation-based prediction models. The proposed framework can support researchers, policymakers, and agricultural practitioners in developing reliable crop yield forecasting systems while improving transparency in model decision-making.

In addition to the main workflow, an independent causal-analysis pipeline was implemented to further examine the relationships between moisture, vegetation, and yield variables. The CCM-based screening highlighted a limited set of meaningful directional

associations, particularly involving soil moisture and vegetation indicators, while the subsequent DoWhy-based analysis provided formal effect estimates under a DAG-based adjustment framework. Among the variables tested, `avg_ssm` showed the largest magnitude effect on yield and was negative, whereas `avg_ndvi` and `avg_fpar` showed positive estimated effects and `avg_rsm` remained close to zero. The accompanying refutation tests remained broadly stable, indicating that these estimates are internally consistent under the adopted model assumptions. Thus, this additional analysis complements the main study by providing an extra layer of result verification and causal interpretation.

This additional pipeline can be extended to several directions in future. First, the CCM stage can be refined by systematically tuning embedding parameters and testing additional variable pairs. Second, the DoWhy stage can be strengthened by exploring alternative estimators and sensitivity analysis. Finally, the current complementary branch for automated preprocessing, causal discovery and robustness evaluation.

Appendices

This appendix contains the supplementary material related to the project, including the references consulted during the study, the GitHub repository containing the implementation, and other supporting resources.

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B. Survey Questionnaire

Not Applicable.

The present study was conducted entirely using publicly available benchmark datasets and secondary agricultural data. Therefore, no survey questionnaire or primary data collection was required.

C. GitHub Repository

The complete implementation of this project, including data preprocessing, exploratory data analysis, causal inference, predictive modelling, explainability analysis, and the Streamlit-based Decision Support System, is available at:

GitHub Repository

<https://github.com/nagsen-aditya2611/Causal-Crop-Yield-Forecasting>

Complementary work for this report is maintained in a separate branch of the repository.

Live Web Application

<https://causal-crop-advisor.streamlit.app/>

D. Additional Documents

The following project documents are included in the GitHub repository or associated cloud storage:

- Final Internship Report
- Source Code
- Processed Dataset

- Figures and Visualizations
- Streamlit Application Files
- Trained Machine Learning and Deep Learning Models